

HSPT Exam Simulator

Full-Length Practice Exam 04

Complete printable test with 298 multiple-choice questions, answer key, and detailed explanations.

Section	Questions	Time
Verbal Skills	60	20 minutes
Quantitative Skills	52	35 minutes
Reading Comprehension	62	35 minutes
Mathematics	64	45 minutes
Language	60	30 minutes

General Directions

Work through each section within the recommended time. Choose the best answer for every question. Mark one answer (A-D) for each item. Use the answer key and explanations only after completing the test.

Recommended total testing time: 165 minutes. Calculators should be used only when permitted by the rules followed for your HSPT preparation.

Section 1: Verbal Skills

60 questions | Recommended time: 20 minutes

1. Choose the best response. Which word is closest in meaning to reluctant?

- A) unwilling
- B) cheerful
- C) exact
- D) temporary

2. Choose the best response. Which word is closest in meaning to preserve?

- A) scatter
- B) protect
- C) predict
- D) exchange

3. Choose the best response. Which word is closest in meaning to evident?

- A) flexible
- B) distant
- C) obvious
- D) complete

4. Choose the best response. Which word is most nearly opposite in meaning to scarce?

- A) uneven
- B) costly
- C) useful
- D) plentiful

5. Choose the best response. Which word is most nearly opposite in meaning to rigid?

- A) flexible
- B) narrow
- C) solid
- D) polished

6. Choose the best response. Which word is most nearly opposite in meaning to conceal?

- A) collect
- B) reveal
- C) repair
- D) observe

7. Choose the best response. Blueprint is to building as recipe is to ____.

- A) kitchen
- B) ingredient
- C) meal
- D) chef

8. Choose the best response. Wool is to sheep as silk is to ____.

- A) thread
- B) moth
- C) fabric
- D) silkworm

9. Choose the best response. Evaporation is to liquid as melting is to ____.

- A) solid
- B) heat
- C) gas
- D) freezing

10. Choose the best response. Which word does not belong with the others?

- A) prairie
- B) harbor
- C) tundra
- D) desert

11. Choose the best response. Which word belongs with sonnet, haiku, and limerick?

- A) canvas
- B) chapter
- C) ballad
- D) speech

12. Choose the best response. Which pair has the same relationship as kennel and dog?

- A) feather and bird
- B) saddle and horse
- C) field and farmer
- D) hive and bee

13. Choose the best response. All objects in cabinet P are metal. No metal object in the workshop is transparent. Which conclusion must be true?

- A) No object in cabinet P is transparent.
- B) Some transparent objects are in cabinet P.
- C) Every metal object is in cabinet P.
- D) Cabinet P itself is metal.

14. Choose the best response. Four exhibits—W, X, Y, and Z—open on consecutive days. X opens before W. Z opens immediately after W. Y opens before X. Which exhibit opens third?

- A) X
- B) W
- C) Y
- D) Z

15. Choose the best response. A code is approved only if it contains a digit and ends with K. Code R contains a digit but does not end with K. What must be true?

- A) Code R begins with K.
- B) Every unapproved code lacks a digit.
- C) Code R is not approved.
- D) Code R contains exactly one digit.

16. Which word is closest in meaning to prudent?

- A) noisy
- B) wise
- C) hasty
- D) fragile

17. Which word is closest in meaning to diminish?

- A) decorate
- B) divide
- C) decrease
- D) discover

18. Which word is closest in meaning to verify?

- A) delay
- B) question
- C) replace
- D) confirm

19. Which word is most nearly opposite in meaning to intricate?

- A) simple
- B) valuable
- C) ancient
- D) colorful

20. Which word is most nearly opposite in meaning to permit?

- A) request
- B) forbid
- C) explain
- D) consider

21. Which word is most nearly opposite in meaning to optimistic?

- A) curious
- B) patient
- C) pessimistic
- D) realistic

22. Compass is to direction as thermometer is to ____.

- A) weather
- B) mercury
- C) degree
- D) temperature

23. Sapling is to tree as calf is to ____.

- A) cow
- B) herd
- C) pasture
- D) hoof

24. Applause is to performance as verdict is to ____.

- A) judge
- B) trial
- C) evidence
- D) courtroom

25. Which word does not belong with the others?

- A) granite
- B) marble
- C) copper
- D) slate

26. Which word belongs with orbit, rotate, and revolve?

- A) wander
- B) settle
- C) stretch
- D) spin

27. Which pair has the same relationship as chapter and book?

- A) scene and play
- B) actor and stage
- C) cover and page
- D) pen and poem

28. A workshop offers ceramics on Tuesday or Thursday, but not both. If ceramics is not offered Thursday, what must be true?

- A) Ceramics is canceled.
- B) Ceramics is offered Tuesday.
- C) Ceramics is offered both days.
- D) No workshop occurs Tuesday.

29. Jules, Kira, and Luis each choose a different task: painting, sorting, or measuring. Jules does not paint. Kira sorts. Luis does not measure. What does Jules do?

- A) painting
- B) sorting
- C) measuring
- D) It cannot be determined.

30. If the gate is locked, the warning light is on. The warning light is off. Which conclusion must be true?

- A) The gate is open to visitors.
- B) The light is broken.
- C) The gate was locked earlier.
- D) The gate is not locked.

31. Which word is closest in meaning to reluctant?

- A) unwilling
- B) cheerful
- C) certain
- D) careless

32. Which word is closest in meaning to scarce?

- A) smooth
- B) ordinary
- C) limited
- D) nearby

33. Which word is closest in meaning to elaborate?

- A) silent
- B) necessary
- C) hidden
- D) detailed

34. Which word is most nearly opposite in meaning to rigid?

- A) flexible
- B) narrow
- C) heavy
- D) polished

35. Which word is most nearly opposite in meaning to conceal?

- A) protect
- B) reveal
- C) collect
- D) remember

36. Which word is most nearly opposite in meaning to abundant?

- A) useful
- B) orderly
- C) meager
- D) uneven

37. Reservoir is to water as granary is to ____.

- A) field
- B) farmer
- C) bread
- D) grain

38. Chisel is to sculptor as stethoscope is to ____.

- A) physician
- B) patient
- C) heartbeat
- D) hospital

39. Whisper is to speak as stroll is to ____.

- A) rest
- B) travel
- C) walk
- D) run

40. Which word is not in the same category as the other three?

- A) lagoon
- B) pond
- C) lake
- D) harbor

41. Which word belongs with cedar, spruce, and fir?

- A) moss
- B) pine
- C) maple
- D) vine

42. Which pair has the same relationship as kitten and litter?

- A) wolf and pack
- B) bird and feather
- C) horse and stable
- D) bee and honey

43. Every manuscript in the cabinet is cataloged. No cataloged item may leave the archive. Which conclusion must be true?

- A) Some manuscripts are not cataloged.
- B) Every archive item is a manuscript.
- C) No manuscript in the cabinet may leave the archive.
- D) The cabinet may leave the archive.

44. Four flags—J, K, L, and M—hang in a row. J is immediately left of K. M is left of J. L is not at either end. Which flag is first?

- A) J
- B) K
- C) L
- D) M

- 45. Exactly two of Ana, Bo, and Cy attend. If Ana attends, Bo does not. If Cy does not attend, Ana attends. Which pair could attend?**
- A) Ana and Bo
 - B) Bo and Cy
 - C) Ana alone
 - D) all three
- 46. Which word is closest in meaning to vigilant?**
- A) weary
 - B) watchful
 - C) generous
 - D) uncertain
- 47. Which word is closest in meaning to impartial?**
- A) fair
 - B) impatient
 - C) silent
 - D) powerful
- 48. Which word is closest in meaning to feasible?**
- A) decorative
 - B) hidden
 - C) possible
 - D) ordinary
- 49. Which word is most nearly opposite in meaning to temporary?**
- A) recent
 - B) fragile
 - C) useful
 - D) permanent
- 50. Which word is most nearly opposite in meaning to timid?**
- A) patient
 - B) bold
 - C) loyal
 - D) calm
- 51. Which word is most nearly opposite in meaning to hinder?**
- A) assist
 - B) observe
 - C) divide
 - D) question
- 52. Blueprint is to building as recipe is to ____.**
- A) kitchen
 - B) ingredient
 - C) meal
 - D) chef

53. Gills are to fish as spiracles are to ____.

- A) mammal
- B) fern
- C) reptile
- D) insect

54. Opaque is to light as impermeable is to ____.

- A) sound
- B) water
- C) heat
- D) stone

55. Three choices are prose forms. Which choice does not belong?

- A) novel
- B) memoir
- C) sonnet
- D) biography

56. Which word belongs with copper, silver, and iron?

- A) glass
- B) granite
- C) clay
- D) tin

57. Which pair has the same relationship as aviary and birds?

- A) apiary and bees
- B) orchard and farmer
- C) aquarium and water
- D) stable and hay

58. All cedar boxes are fragrant. No fragrant object is stored in Room K. Which statement must be true?

- A) Every box outside Room K is cedar.
- B) Some fragrant objects are cedar boxes.
- C) No cedar box is stored in Room K.
- D) Room K contains no boxes.

59. Mara, Niko, Oren, and Pia present one at a time. Oren presents after Mara but before Pia. Niko presents after Pia. Who presents second?

- A) Mara
- B) Niko
- C) Pia
- D) Oren

60. A code has three different digits chosen from 2, 4, 6, and 8. The first digit is twice the second. The last digit is greater than the first. Which code works?

- A) 248
- B) 426
- C) 624
- D) 842

Section 2: Quantitative Skills

52 questions | Recommended time: 35 minutes

61. Choose the best response. What number comes next? 90, 81, 69, 54, 36, ____

- A) 15
- B) 12
- C) 18
- D) 21

62. Choose the best response. What number comes next? 4, 12, 6, 18, 9, 27, ____

- A) 12
- B) 13.5
- C) 18
- D) 40.5

63. Choose the best response. What number comes next? 1, 4, 10, 19, 31, ____

- A) 43
- B) 45
- C) 46
- D) 49

64. Choose the best response. What number comes next? 2, 3, 6, 11, 18, ____

- A) 25
- B) 28
- C) 29
- D) 27

65. Choose the best response. A price after a 20% discount is \$68. What was the original price?

- A) \$85
- B) \$81.60
- C) \$88
- D) \$136

66. Choose the best response. When a number is divided by 7, the quotient is 8 and the remainder is 5. What is the number?

- A) 51
- B) 61
- C) 56
- D) 91

67. Choose the best response. The ratio of red beads to white beads is 5:3. If there are 32 beads total, how many are white?

- A) 8
- B) 12
- C) 15
- D) 20

68. Choose the best response. Quantity A: $\frac{5}{6} - \frac{1}{4}$. Quantity B: $\frac{7}{12}$.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

69. Choose the best response. n is an integer greater than 4. Quantity A: $3n$. Quantity B: $n + 8$.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

70. Choose the best response. p and q are positive, and $pq = 24$. Quantity A: $p + q$. Quantity B: 10.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

71. Choose the best response. A circle has diameter 14. Quantity A: its radius. Quantity B: 7.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

72. Choose the best response. A regular hexagon has side length 4. Quantity A: its perimeter. Quantity B: 25.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

73. Choose the best response. A rectangle has perimeter 30 and length x . Quantity A: its width. Quantity B: 6.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

74. What number comes next? 160, 80, 40, 20, ____

- A) 10
- B) 5
- C) 15
- D) 30

75. What number comes next? 3, 7, 15, 31, 63, ____

- A) 95
- B) 126
- C) 127
- D) 129

76. What number comes next? 2, 5, 4, 10, 8, 20, ____

- A) 18
- B) 30
- C) 40
- D) 16

77. Which value of k makes $5(k - 2) = 35$ true?

- A) 9
- B) 5
- C) 7
- D) 11

78. What is the least positive number divisible by 6, 8, and 10?

- A) 24
- B) 80
- C) 120
- D) 240

79. A recipe uses flour and oats in a 7:2 ratio. If it uses $3\frac{1}{2}$ cups of flour, how many cups of oats are needed?

- A) $\frac{1}{2}$ cup
- B) $1\frac{1}{2}$ cups
- C) 2 cups
- D) 1 cup

80. Which expression is equivalent to 0.06×250 ?

- A) 15
- B) 1.5
- C) 150
- D) 1,500

81. Quantity A: $2^3 + 3^2$. Quantity B: 18.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

82. x is a negative number. Quantity A: $x - 4$. Quantity B: $x + 4$.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

83. a is a nonzero real number. Quantity A: a/a . Quantity B: 1.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

84. A triangle has side lengths 5, 5, and 6. Quantity A: its perimeter. Quantity B: 17.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

85. A rectangle has area 48 square units and length 8 units. Quantity A: its width. Quantity B: 6 units.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

86. A circle has radius r , where $r > 0$. Quantity A: its diameter. Quantity B: $r + 5$.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

87. What number comes next? 4, 9, 17, 28, 42, ____

- A) 59
- B) 54
- C) 56
- D) 60

88. What number comes next? 81, 27, 30, 10, 13, ____

- A) 3
- B) $13/3$
- C) $10/3$
- D) 16

89. What number comes next? 1, 4, 3, 12, 5, 20, 7, ____

- A) 24
- B) 27
- C) 28
- D) 35

90. What number comes next? 2, 6, 24, 120, ____

- A) 240
- B) 600
- C) 840
- D) 720

91. If $\frac{3}{5}$ of a number is 42, what is the number?

- A) 70
- B) 25.2
- C) 63
- D) 105

92. Which number leaves a remainder of 2 when divided by both 5 and 7?

- A) 37
- B) 52
- C) 77
- D) 72

93. A price rises from \$48 to \$60. What is the percent increase?

- A) 12%
- B) 20%
- C) 25%
- D) 80%

94. n is an integer greater than 1. Quantity A: the number of positive factors of n . Quantity B: 2.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

95. Quantity A: $|-18| - 7$. Quantity B: 3^2 .

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

96. $p + q = 20$. Quantity A: $p + q - 6$. Quantity B: 14.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

97. A regular hexagon has side length 4. Quantity A: its perimeter. Quantity B: 25.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

98. Two angles are supplementary, and one measures x° . Quantity A: the other angle. Quantity B: 90° .

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

99. A square has diagonal length 10. Quantity A: one side length. Quantity B: 8.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

100. What number comes next? 3, 8, 18, 38, 78, ____

- A) 158
- B) 118
- C) 148
- D) 156

101. What number comes next? 96, 48, 52, 26, 30, ____

- A) 13
- B) 17
- C) 15
- D) 60

102. What number comes next? 2, 5, 4, 10, 8, 20, 16, ____

- A) 24
- B) 32
- C) 36
- D) 40

103. What number comes next? 7, 10, 16, 25, 37, ____

- A) 52
- B) 49
- C) 50
- D) 55

104. A number is increased by 25% and becomes 90. What was the original number?

- A) 67.5
- B) 75
- C) 72
- D) 112.5

105. When an integer is divided by 8, the quotient is 6 and the remainder is 5. What is the integer?

- A) 43
- B) 48
- C) 61
- D) 53

106. The ratio of blue tiles to white tiles is 5:3. If there are 64 tiles, how many are white?

- A) 24
- B) 21
- C) 32
- D) 40

107. x is a negative number. Quantity A: x^3 . Quantity B: x^2 .

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

108. $a - b = 9$. Quantity A: $a + 4$. Quantity B: $b + 13$.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

109. r and s are positive integers with $rs = 24$. Quantity A: $r + s$. Quantity B: 11.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

110. A rectangle has length 9 and width 5. Quantity A: its area. Quantity B: its perimeter.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

111. An isosceles triangle has two angles of 68° each. Quantity A: the third angle. Quantity B: 45° .

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

112. A cube has edge length 3. Quantity A: its volume. Quantity B: its total surface area.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

Section 3: Reading Comprehension

62 questions | Recommended time: 35 minutes

Literary Passage: The Last Clay Whistle

Nico had promised to help pack his aunt's pottery studio before the building changed owners. Most shelves emptied quickly, but one held dozens of small clay whistles shaped like birds. Years earlier, Aunt Leena had sold them at every spring market. Now several were chipped, and none had price tags.

Nico suggested wrapping them all. His aunt shook her head. Moving boxes were nearly full, and the new studio had less storage. She selected one blue whistle, tested its clear note, and placed it beside the tape. The rest, she said, could become fresh clay if soaked and worked again.

The idea startled Nico. He had treated each bird as a finished memory. Aunt Leena treated the clay as material still capable of change. Together they removed bits of glaze and lowered the plain whistles into a bucket. The sharp outlines slowly softened beneath the water.

Before leaving, Nico wrapped the blue bird in newspaper. He no longer saw the bucket as a container of ruined keepsakes. It held tomorrow's bowls, tiles, or perhaps another flock of whistles—forms no one had imagined yet.

113. Choose the best response. Why can Aunt Leena not keep every whistle?

- A) Her moving boxes are nearly full, and the new studio has less storage.
- B) The whistles belong to the building owner.
- C) Nico refuses to wrap them.
- D) The market has forbidden their sale.

114. Choose the best response. Why is Nico initially startled by soaking the whistles?

- A) He believes water will make them whistle louder.
- B) He thinks of them as completed memories rather than reusable material.
- C) He plans to sell every bird that day.
- D) He has never seen clay before.

115. Choose the best response. As used in paragraph 3, plain most nearly means ____.

- A) ordinary in shape
- B) easy to understand
- C) unglazed
- D) unattractive

116. Choose the best response. The bucket of softened clay mainly symbolizes ____.

- A) the danger of forgetting family promises
- B) the high cost of moving supplies
- C) the end of Aunt Leena's career
- D) the possibility of creating something new from the past

117. Choose the best response. Which statement best expresses the story's central idea?

- A) Letting an object change can preserve its value in a new form.
- B) Every handmade object should be stored forever.
- C) Moving is easier when relatives work alone.
- D) Artists should create only useful objects.

Informational Passage: Why Some Seeds Wait

A seed may appear inactive, yet it contains a living embryo and a supply of stored food. Many seeds begin growing when moisture, oxygen, and a suitable temperature occur together. Others remain dormant even when these basic conditions seem favorable.

Dormancy can prevent a seed from sprouting at a dangerous time. Some desert plants produce seeds with coatings that water cannot easily enter. Months of weathering may weaken the coating, delaying growth until rainfall is more likely to continue. Certain forest seeds respond to a period of winter cold. This requirement helps keep them from sprouting during a brief warm spell in autumn.

Fire can also provide a signal. In several plant species, heat cracks a hard seed coat, while chemicals in smoke trigger changes inside the seed. Such adaptations allow new plants to emerge after flames have opened sunny space and returned nutrients to the soil.

Dormancy is therefore not simply a failure to grow. It is a timing strategy. By responding to particular environmental cues, seeds improve the chance that fragile seedlings will encounter conditions they can survive.

118. Choose the best response. Which three basic conditions help many seeds begin growing?

- A) Smoke, darkness, and salt
- B) Moisture, oxygen, and suitable temperature
- C) Cold, wind, and deep soil
- D) Sunlight, insects, and fertilizer

119. Choose the best response. How can winter cold help certain forest seeds?

- A) It permanently hardens their coats.
- B) It supplies stored food to embryos.
- C) It prevents sprouting during a brief autumn warm spell.
- D) It removes all moisture from the soil.

120. Choose the best response. Why might fire-triggered seeds benefit after a fire?

- A) Their embryos no longer need oxygen.
- B) Their roots can grow without water.
- C) Their coats become completely fireproof.
- D) They face more sunlight and nutrients with less competition.

121. Choose the best response. As used in the last paragraph, cues most nearly means ____.

- A) signals
- B) barriers
- C) supplies
- D) measurements

122. Choose the best response. How is the passage mainly organized?

- A) It compares two experiments in chronological order.
- B) It explains dormancy, gives examples of triggers, and states its survival advantage.
- C) It lists seeds from smallest to largest.
- D) It argues that all seeds require fire.

Reflective Passage: The Value of a Draft

A first draft is often judged by what it lacks. Its opening may wander, its evidence may be thin, and its conclusion may arrive too quickly. Because the finished work looks smoother, the rough version can seem like an embarrassing stage best hidden from view.

Yet a draft performs a different job from a final piece. It makes thought visible. Once an idea exists on a page, a writer can test its order, notice gaps, and discover connections that silent planning did not reveal. Crossed-out sentences are not proof that writing failed; they record decisions.

This principle reaches beyond essays. An engineer's early model exposes weaknesses before expensive materials are used. A musician's rehearsal uncovers an awkward transition. In each case, the unfinished version creates a safe place for correction.

Improvement still requires discipline. Repeating the same attempt without reflection is not revision. The maker must compare the result with the goal and decide what to change. A draft has value not because imperfection is automatically admirable, but because visible imperfection gives purposeful work somewhere to begin.

123. Choose the best response. According to the passage, what can an engineer's early model reveal?

- A) The exact price of every material
- B) Whether an essay needs more evidence
- C) Weaknesses before costly materials are used
- D) How an audience will applaud

124. Choose the best response. What is the author's central claim?

- A) A polished result should display every earlier mistake.
- B) Planning silently is always better than drafting.
- C) Imperfection alone guarantees creativity.
- D) Unfinished work is valuable when it makes problems visible for thoughtful revision.

125. Choose the best response. What does the author imply about crossing out a sentence?

- A) It may show that a writer evaluated and improved an idea.
- B) It proves the writer had no plan.
- C) It should be avoided in every first draft.
- D) It makes the final piece less organized.

126. Choose the best response. The examples of an engineer and a musician primarily serve as ____.

- A) statistics proving that drafts save money
- B) analogies extending the idea beyond writing
- C) contrasts showing that writing is easier
- D) definitions of technical vocabulary

127. Choose the best response. Which word best describes the author's tone?

- A) sarcastic
- B) fearful
- C) reasoned
- D) nostalgic

128. Choose the best response. As used in paragraph 2, thin most nearly means ____.

- A) narrow in width
- B) quietly spoken
- C) widely scattered
- D) insufficient

Literary Passage: The Quiet Rehearsal

Mara had practiced her cello solo for weeks, but the auditorium felt different when it was empty. Every scrape of a chair echoed, and the stage lights erased the audience seats beyond the first row. During the dress rehearsal, she rushed the opening and missed a shift she had played correctly at home.

Mr. Ibarra did not ask her to begin again. Instead, he walked to the back wall and asked Mara to play one sustained note toward him. She drew the bow slowly. The note traveled through the room and returned as a faint

reflection. Then he moved to the balcony and asked for the same note.

By testing the hall, Mara began listening outward instead of only monitoring her fingers. She noticed that quick passages blurred when she hurried, while rests seemed shorter under the bright lights. On her next run, she allowed each phrase to reach the room before beginning the next.

The mistake in the opening did not disappear from her memory. It became useful information. When the audience arrived that evening, the hall was no longer an unfamiliar space waiting to expose her errors. It was part of the instrument she had learned to play.

129. What causes Mara difficulty during the dress rehearsal?

- A) A string breaks during the solo.
- B) The auditorium is filled with talking students.
- C) Mr. Ibarra changes the music.
- D) She rushes the opening and misses a shift.

130. Why does Mr. Ibarra ask Mara to sustain one note from different places?

- A) He wants her to learn how sound behaves in the auditorium.
- B) He is testing whether she remembers the melody.
- C) He wants to shorten the concert.
- D) He cannot hear notes from the stage.

131. Calling the hall “part of the instrument” is an example of ____.

- A) rhyme
- B) metaphor
- C) understatement
- D) flashback

132. Which statement best expresses the story’s central idea?

- A) Musicians should practice only in empty rooms.
- B) A perfect rehearsal guarantees a perfect concert.
- C) Careful attention can turn an unfamiliar setting and a mistake into useful guidance.
- D) Bright lights improve musical accuracy.

133. How does Mara’s attitude toward the auditorium change?

- A) She first sees it as unsettling but later understands how to work with it.
- B) She first admires it but later refuses to enter it.
- C) She remains unaware of it throughout rehearsal.
- D) She first finds it too small but later finds it crowded.

Scientific Passage: A Map Made by Sound

Bats that hunt at night cannot always rely on vision. Many species use echolocation: they release high-frequency calls and interpret the returning echoes. An echo's delay indicates how far away an object is. Changes in loudness and pitch can provide clues about the object's size, texture, direction, and motion.

Producing a call is only the beginning. A bat must separate useful echoes from sounds made by wind, insects, other bats, and its own movements. Its ears and nervous system process these signals extremely quickly. As a bat approaches an insect, it often sends calls more rapidly, receiving updated information at shorter intervals. Different surroundings favor different call patterns. In open air, longer calls can help detect distant targets. Near branches and leaves, shorter calls reduce overlap between outgoing sounds and returning echoes. Even bats of the same species may adjust their calls as conditions change.

Echolocation is sometimes described as seeing with sound, but the comparison is incomplete. It is an active process: the bat selects a call, receives a response, and changes its next call based on what it has learned.

134. What information does an echo's delay provide?

- A) the object's exact color
- B) the distance to an object
- C) the bat's age
- D) the temperature of the air

135. As used in paragraph 2, separate most nearly means ____.

- A) divide physically
- B) send away
- C) distinguish
- D) measure equally

136. Why do bats often call more rapidly as they approach an insect?

- A) Rapid calls make the insect stop moving.
- B) Their vision becomes stronger at close range.
- C) Long calls cannot travel through air.
- D) They need frequent updates about a nearby moving target.

137. What is the passage mainly explaining?

- A) How bats actively adjust echolocation to gather information in different conditions
- B) Why every bat uses identical calls
- C) How insects imitate bat sounds
- D) Why nighttime air blocks vision completely

138. Why does the author discuss "seeing with sound" in the final paragraph?

- A) To prove that bats see colors through echoes
- B) To introduce a familiar comparison and then clarify its limitation
- C) To argue that vision and hearing are identical
- D) To provide a humorous ending

Informational Passage: When a City Uncovers a Stream

As cities expanded, some small streams were straightened, covered, or directed into underground pipes. This created land for streets and buildings and moved storm water away quickly. Over time, however, buried channels sometimes became too narrow for heavier runoff, and residents lost access to natural waterways.

A process called daylighting brings a covered stream back to the surface. Engineers first study old maps, current drainage patterns, soil, and nearby structures. The restored channel may not follow its exact historical route because modern buildings and utilities occupy the land. Instead, designers seek a workable path that safely carries water.

A daylighted stream can provide habitat, cool nearby air, and create public green space. Curves, plants, and areas that temporarily hold water may also slow runoff. Yet projects require careful maintenance. Trash can block the channel, newly planted banks may erode, and poorly designed paths can place visitors too close to fast water.

Daylighting is therefore neither a simple return to the past nor a universal solution. It is a redesign that combines evidence about an older landscape with the needs and limits of a modern city.

139. What do engineers study before daylighting a stream?

- A) Only recent rainfall totals
- B) Fish populations in distant rivers
- C) Old maps, drainage patterns, soil, and nearby structures
- D) The prices of surrounding homes alone

140. Why may a restored channel differ from its historical route?

- A) Old maps never show waterways.
- B) Curved streams cannot carry water.
- C) Engineers always prefer underground pipes.
- D) Modern buildings and utilities may occupy the original path.

141. How can curves and planted holding areas reduce flooding risk?

- A) They can slow runoff instead of sending it away immediately.
- B) They prevent all rain from reaching the ground.
- C) They make underground pipes narrower.
- D) They permanently remove water from the city.

142. As used in paragraph 3, banks most nearly means ____.

- A) financial institutions
- B) land along the sides of a stream
- C) rows of switches
- D) stored supplies

143. Which statement best expresses the author's main point?

- A) Every buried stream should follow its exact original route.
- B) City growth makes natural waterways unnecessary.
- C) Daylighting adapts an old waterway to modern conditions and requires careful design.
- D) A restored stream requires no maintenance.

144. How is the passage mainly organized?

- A) It presents instructions for building a pipe.
- B) It compares streams solely by length.
- C) It narrates one resident's childhood memory.
- D) It describes a past practice, explains a redesign process, and weighs benefits and challenges.

Literary Passage: The Borrowed Bell

Nico found the brass handbell in a cabinet while helping prepare the old meetinghouse for its reopening. Its wooden handle was worn smooth, and a paper tag read, 'School bell, 1912.' He pictured a teacher ringing it above a noisy room.

The event coordinator asked Nico to place it in a display case, but the case's latch was broken. Afraid someone might knock the bell down, he carried it to the office. On the way, he heard visitors entering early. The coordinator, across the hall, pointed urgently toward the front steps.

Nico assumed she wanted help directing the crowd. Then he noticed the new electric chime had failed. He raised the old bell and rang once. Its clear note traveled through the hallway, and conversations softened. He rang again, and the visitors gathered.

Afterward, the bell went into the repaired case. Nico's imagined history was no longer all he knew about it. For one brief morning, the object had performed its old task again.

145. Why does Nico carry the bell to the office?

- A) The display case has a broken latch.
- B) He wants to polish the handle.
- C) A visitor asks to borrow it.
- D) The tag lists the wrong date.

146. What does the coordinator most likely mean by pointing toward the steps?

- A) She wants him to lock the meetinghouse.
- B) She wants Nico to use the bell to gather arriving visitors.
- C) She wants him to replace the paper tag.
- D) She wants him to repair the case immediately.

147. What is the effect of the bell performing its old task again?

- A) It proves modern equipment is never useful.
- B) It shows the bell belongs to Nico.
- C) It connects the building's history with its reopening.
- D) It explains how the bell was manufactured.

148. As used in paragraph 3, softened most nearly means ____.

- A) turned warmer
- B) grew kinder
- C) lost detail
- D) became quieter

149. What central idea is developed through Nico's experience?

- A) An old object can gain fresh meaning when it serves a present need.
- B) Valuable objects should never be displayed.
- C) Early visitors usually cause confusion.
- D) Electronic devices are difficult to repair.

Literary Passage: A Line Across the Floor

The robotics team had taped a crooked line across the workshop floor to test whether their small rover could follow a route. Tessa had written the steering code, and during three trials the rover rolled past the sharpest turn. Each failure felt to her like a public correction written in blinking red lights. Dev examined the wheels instead of the screen. He placed a ruler beside them and discovered that the left wheel was slightly larger because its rubber tread had stretched. Tessa almost objected; the program, she thought, should be able to overcome imperfect hardware. They changed one value so the left motor turned a little more slowly. This time the rover followed the tape until a loose corner folded over. It stopped, uncertain, at the blank patch. Tessa laughed first. She pressed the tape flat and added a note to the team's report: 'A system includes everything it depends on.'

The rover's successful final run mattered, but so did the sentence. It reminded the team to examine code, machines, and surroundings together.

150. What physical problem does Dev discover?

- A) The battery is nearly empty.
- B) The left wheel is slightly larger than the right.
- C) The ruler is bent.
- D) The floor is sloped.

151. Why does Tessa almost object to Dev's discovery?

- A) She does not want the rover to finish.
- B) She thinks the tape should be removed.
- C) She believes her code should compensate for imperfect equipment.
- D) She doubts that the wheels can turn.

152. How is the story mainly organized?

- A) A narrator compares two unrelated inventions.
- B) Instructions are listed from easiest to hardest.
- C) Events are told backward from the final run.
- D) A repeated failure leads to one discovery, then another, and finally a broader lesson.

153. The phrase “a public correction written in blinking red lights” mainly emphasizes ____.

- A) Tessa's embarrassment about the failures
- B) the brightness of the workshop
- C) the rover's ability to write
- D) Dev's dislike of the program

154. What lesson does the team learn?

- A) A rover works best without written code.
- B) Successful design requires considering software, hardware, and surroundings together.
- C) Rubber wheels should always be identical in color.
- D) Testing should stop after the first success.

Informational Passage: Why Museums Rotate Displays

A museum may own thousands of objects while showing only a small fraction of them. Limited gallery space is one reason, but preservation also matters. Light can fade pigments and weaken paper or fabric. Changes in humidity may cause wood to expand and contract, while dust and handling can damage delicate surfaces. Rotating displays gives sensitive objects periods of rest in controlled storage. During that time, specialists can inspect them, record changes, and make careful repairs. Rotation also allows curators to build new exhibits around different questions. The same bowl, letter, or tool may help tell several stories depending on the objects placed beside it.

Storage is not simply a dark warehouse. Well-managed collections use labeled locations, stable environmental conditions, and detailed records. Researchers may study stored objects even when the public cannot see them. Some museums also publish digital images and descriptions, increasing access without exposing an object to additional light or handling.

Thus, an absent object is not necessarily forgotten. Its time away from the gallery can protect it and prepare it to contribute to a future exhibit.

155. According to the passage, how can humidity harm wooden objects?

- A) It can erase labels instantly.
- B) It can turn wood into paper.
- C) It can make the wood expand and contract.
- D) It can make galleries smaller.

156. As used in paragraph 2, sensitive most nearly means ____.

- A) able to feel emotion
- B) secret
- C) accurately measured
- D) easily damaged

157. Why might a curator place the same object in different exhibits?

- A) Its meaning can change with the question and surrounding objects.
- B) The museum has no other objects.
- C) Its materials change each time it is moved.
- D) Researchers refuse to study stored items.

158. Which statement best captures the purpose of the museum passage?

- A) Why museums should replace objects with copies
- B) Why keeping some museum objects off display can preserve them and support future use
- C) How visitors choose which galleries to enter
- D) Why storage buildings must be open to everyone

159. Which best describes the author's approach?

- A) The author argues that all collections are too large.
- B) The author narrates the discovery of one ancient object.
- C) The author gives step-by-step repair directions.
- D) The author corrects a possible misconception by explaining several purposes of storage.

Literary Passage: The Unmarked Trail

At the edge of the community garden, Amaya noticed a narrow strip of flattened grass leading behind the tool shed. The official paths were covered with wood chips, so she assumed the strip was a shortcut made by impatient visitors. She planned to ask the garden manager to block it.

The next morning, Amaya arrived early to water the seedlings. A rabbit emerged from the strip, paused beside the rain barrel, and disappeared beneath the fence. Soon a pair of sparrows hopped along the same route, gathering dry grass. Amaya crouched quietly and saw tiny tracks pressed into the soil.

Instead of placing a barrier there, she set two small stones beside the strip so gardeners would notice and avoid stepping on it. She also made a sign for the shed: "Wildlife crossing—please leave clear."

By midsummer, taller plants partly hid the trail. Visitors sometimes asked where it led. Amaya would explain that it was not really theirs to follow. The garden, she had learned, contained paths made for purposes other than human convenience.

160. Why does Amaya initially want the strip blocked?

- A) She sees rabbits eating every seedling.
- B) She thinks visitors created an improper shortcut.
- C) The manager orders her to close it.
- D) The strip leads outside the garden.

161. Why does Amaya place stones beside the strip?

- A) To mark where a new shed will stand
- B) To prevent rainwater from reaching the plants
- C) To help people notice the route without obstructing animals
- D) To guide visitors toward the fence

162. What does the unmarked trail mainly symbolize?

- A) Amaya's wish to leave the garden
- B) The manager's poor planning
- C) A competition among gardeners
- D) The needs and activity that humans may overlook

163. As used in the final paragraph, convenience most nearly means ____.

- A) ease or usefulness
- B) ownership
- C) curiosity
- D) safety

164. Which theme is best supported by the story?

- A) Rules should never be changed.
- B) Careful observation can challenge a quick assumption.
- C) Wild animals cannot live near people.
- D) Gardens need more marked paths.

Literary Passage: One Measure Short

Eli had promised to bring enough lemonade for the neighborhood concert. His grandmother's recipe filled one large pitcher, but he needed three. He multiplied every amount carefully until he reached the honey: the jar held one cup less than his calculation required.

His first thought was to make only two pitchers and hope no one noticed. Then he tasted a spoonful of the tart lemon mixture. Adding less honey to all three pitchers would make the drink unpleasantly sour.

Eli searched the kitchen and found ripe peaches. He simmered them with the remaining honey, strained the syrup, and divided it among the pitchers. The result tasted different from his grandmother's lemonade—so different that Eli hesitated to serve it under her handwritten label. He crossed out "Grandma's Lemonade" and wrote "Peach-Lemon Trial" instead.

At the concert, the first pitcher emptied quickly. His grandmother took a cup from the second and studied it before smiling. "You changed the recipe," she said.

"I ran out of honey."

"No," she replied, lifting her cup. "You ran into an idea."

165. What shortage does Eli face?

- A) He has no lemons.
- B) He needs a fourth pitcher.
- C) He has one cup less honey than he needs.
- D) The peaches are unripe.

166. Why does Eli change the label?

- A) His grandmother forbids him to use her name.
- B) He expects the experiment to fail.
- C) The old label will not stick to the pitcher.
- D) He wants the name to honestly reflect his altered drink.

167. How does the shortage affect the plot?

- A) It creates a problem that leads Eli to invent a new recipe.
- B) It causes the concert to be canceled.
- C) It proves the original recipe was incorrect.
- D) It makes Eli abandon the lemonade project.

168. What does the grandmother mean by “You ran into an idea”?

- A) Eli spilled the drink while running.
- B) The shortage unexpectedly inspired a creative solution.
- C) The recipe was hidden inside the jar.
- D) Eli should have followed the recipe exactly.

169. Which statement best expresses a central idea of the story?

- A) Popular recipes require expensive ingredients.
- B) Promises should be avoided when supplies are low.
- C) Adapting honestly can turn a limitation into an opportunity.
- D) Family recipes must never be changed.

Informational Passage: How Seed Banks Protect Variety

Farmers and gardeners depend on plants that can thrive under particular conditions. Yet a crop variety may disappear when people stop growing it, when disease spreads, or when its habitat changes. Seed banks help preserve plant diversity by storing seeds from many varieties and wild relatives of crops.

Workers first clean and dry collected seeds. Lower moisture slows the chemical activity that causes seeds to age. The seeds are sealed in labeled containers and kept cold. Records identify where and when each sample was collected and describe important traits.

Storage does not keep every seed alive forever. Staff periodically remove a small sample and test whether the seeds still sprout. If too few germinate, workers grow some remaining seeds under controlled conditions, collect fresh seeds, and return the new generation to storage.

Seed banks are not replacements for farms or natural habitats. Plants continue to evolve only when they grow and reproduce in changing environments. Instead, stored collections provide a backup. Researchers may use them to study plant traits, and breeders may find useful characteristics, such as resistance to a disease. By protecting many kinds of seeds, banks preserve options that future communities may need.

170. Why are seeds dried before cold storage?

- A) Dry seeds become larger.
- B) Moisture removes useful traits.
- C) Drying guarantees permanent survival.
- D) Reducing moisture slows the processes that age them.

171. As used in paragraph 3, germinate most nearly means ____.

- A) begin to grow
- B) become dry
- C) resist disease
- D) remain frozen

172. Why do seed-bank records include where a sample was collected?

- A) The location determines the container's color.
- B) The origin may help researchers understand its traits and conditions.
- C) Only farmers from that place may use it.
- D) Seeds must be returned yearly.

173. What is the passage mainly about?

- A) Why all crops should be grown indoors
- B) How cold temperatures create new species
- C) How seed banks store, maintain, and make plant diversity available
- D) Why wild plants are replacing farm crops

174. Why does the author mention that seed banks cannot replace habitats?

- A) To argue that storing seeds is pointless
- B) To show that farms prevent evolution
- C) To introduce instructions for building a habitat
- D) To clarify both the value and the limits of stored collections

Section 4: Mathematics

64 questions | Recommended time: 45 minutes

175. Choose the best response. What is the value of $|-17| - |6 - 14|$?

- A) 9
- B) -25
- C) 11
- D) 25

176. Choose the best response. Which number is greatest?

- A) 0.72
- B) $11/15$
- C) 71%
- D) $7/10$

177. Choose the best response. What is the greatest common factor of 84 and 126?

- A) 14
- B) 21
- C) 42
- D) 252

178. Choose the best response. A jar is $\frac{3}{5}$ full. After 18 ounces are added, it is $\frac{9}{10}$ full. What is the jar's capacity?

- A) 30 ounces
- B) 45 ounces
- C) 90 ounces
- D) 60 ounces

179. Choose the best response. Solve $5 - 2(3x - 4) = -17$.

- A) 5
- B) 3
- C) 4
- D) 6

180. Choose the best response. If $f(t) = t^2 - 3t$, what is $f(-2)$?

- A) -10
- B) 10
- C) 2
- D) 14

181. Choose the best response. A line follows $y = -2x + 7$. If $y = -5$, what is x ?

- A) -6
- B) 1
- C) 6
- D) 12

182. Choose the best response. The length of a rectangle is 3 more than twice its width. Its perimeter is 42. What is the width?

- A) 5
- B) 7
- C) 9
- D) 6

183. Choose the best response. Each exterior angle of a regular polygon measures 45° . How many sides does the polygon have?

- A) 8
- B) 6
- C) 10
- D) 12

184. Choose the best response. A circle has circumference 31.4 units. Using 3.14 for π , what is its diameter?

- A) 5 units
- B) 10 units
- C) 15.7 units
- D) 20 units

185. Choose the best response. A rectangular prism measures 3 by 4 by 8. What is its surface area?

- A) 96 square units
- B) 112 square units
- C) 136 square units
- D) 192 square units

186. Choose the best response. A cyclist travels 18 miles in 1 hour 12 minutes. What is the average speed?

- A) 12 miles per hour
- B) 18 miles per hour
- C) 21.6 miles per hour
- D) 15 miles per hour

187. Choose the best response. A package has a mass of 2.35 kilograms. What is its mass in grams?

- A) 2,350 grams
- B) 23.5 grams
- C) 235 grams
- D) 23,500 grams

188. Choose the best response. The values 4, 7, 7, 9, 13, and 14 are recorded. What is the range minus the mode?

- A) -3
- B) 3
- C) 7
- D) 10

189. Choose the best response. A spinner has eight equal sections numbered 1 through 8. What is the probability of landing on a prime number?

- A) $\frac{3}{8}$
- B) $\frac{5}{8}$
- C) $\frac{1}{2}$
- D) $\frac{3}{4}$

190. Choose the best response. A class survey reports 12 walk, 9 bus, 6 bicycle, and 3 car trips. If one response is chosen at random, what is the probability it is bus or bicycle?

- A) $\frac{2}{5}$
- B) $\frac{3}{5}$
- C) $\frac{5}{6}$
- D) $\frac{1}{2}$

191. What is $2\frac{3}{4} \div 1\frac{1}{10}$?

- A) $1\frac{3}{5}$
- B) $2\frac{1}{2}$
- C) $2\frac{1}{4}$
- D) $3\frac{1}{4}$

192. Which expression has a value of 36?

- A) $72 \div 3 + 6$
- B) $5^2 + 6$
- C) $3(18 - 6)$
- D) $4 \times 8 - 2$

193. A number is 35% of 240. What is 25% of that number?

- A) 30
- B) 54
- C) 84
- D) 21

194. Simplify $4(2y - 3) - 5y$.

- A) $3y - 12$
- B) $3y - 3$
- C) $13y - 12$
- D) $8y - 15$

195. Solve $(x + 5)/3 = 8$.

- A) 11
- B) 19
- C) 21
- D) 29

196. Which inequality represents “seven less than twice n is at most 15”?

- A) $2(n - 7) \geq 15$
- B) $7 - 2n < 15$
- C) $2n - 7 \leq 15$
- D) $2n + 7 \leq 15$

197. A taxi charges \$4 plus \$2.50 per mile. Which expression gives the cost for m miles?

- A) $6.5m$
- B) $4m + 2.5$
- C) $2.5(m + 4)$
- D) $4 + 2.5m$

198. Two angles of a triangle measure 38° and 67° . What is the third angle?

- A) 75°
- B) 65°
- C) 85°
- D) 105°

199. A right triangle has legs 9 and 12. What is the hypotenuse?

- A) 13
- B) 15
- C) 18
- D) 21

200. A parallelogram has base 11 units and height 7 units. What is its area?

- A) 36 square units
- B) 63 square units
- C) 77 square units
- D) 154 square units

201. A map scale is 1 inch = 18 miles. Two towns are $3\frac{1}{2}$ inches apart. What is their actual distance?

- A) 54 miles
- B) 72 miles
- C) 81 miles
- D) 63 miles

202. A tank contains 5 gallons 2 quarts. How many quarts is that?

- A) 22 quarts
- B) 12 quarts
- C) 20 quarts
- D) 28 quarts

203. A movie begins at 6:47 p.m. and lasts 2 hours 26 minutes. When does it end?

- A) 8:53 p.m.
- B) 9:13 p.m.
- C) 9:03 p.m.
- D) 9:23 p.m.

204. The mean of five scores is 82. Four scores are 75, 80, 84, and 86. What is the fifth score?

- A) 81
- B) 83
- C) 85
- D) 87

205. A bag contains 4 green, 3 yellow, and 5 purple tiles. What is the probability of selecting a tile that is not purple?

- A) $\frac{5}{12}$
- B) $\frac{1}{2}$
- C) $\frac{7}{5}$
- D) $\frac{7}{12}$

206. Data set A has range 12. Data set B has minimum 9 and maximum 24. How much greater is B's range than A's?

- A) 3
- B) 2
- C) 12
- D) 15

207. Which fraction is halfway between $\frac{2}{3}$ and $\frac{5}{6}$?

- A) $\frac{3}{4}$
- B) $\frac{7}{12}$
- C) $\frac{4}{5}$
- D) $\frac{11}{12}$

208. What is the greatest common factor of 84 and 126?

- A) 14
- B) 42
- C) 21
- D) 252

209. Which value is greatest?

- A) 57%
- B) 0.58
- C) $\frac{7}{12}$
- D) 0.579

210. Evaluate $6 + 24 \div (3 \times 2) - 1$.

- A) 4
- B) 15
- C) 29
- D) 9

211. If $f(x) = 3x^2 - 2$, what is $f(-2)$?

- A) 10
- B) -14
- C) 8
- D) 34

212. Solve $4(x - 3) = 2x + 10$.

- A) 5
- B) 11
- C) 8
- D) 16

213. The first term of a pattern is 7, and each new term is 4 more than the previous term. Which expression gives term n ?

- A) $4n + 7$
- B) $7n - 4$
- C) $4n + 3$
- D) $3n + 4$

214. A club pays a one-time fee of \$90 and \$18 for each participant. Its budget is at most \$360. What is the greatest number of participants?

- A) 12
- B) 16
- C) 20
- D) 15

215. Each exterior angle of a regular polygon measures 45° . How many sides does it have?

- A) 8
- B) 6
- C) 10
- D) 12

216. A rectangular prism measures 3 by 4 by 8 units. What is its surface area?

- A) 96 square units
- B) 136 square units
- C) 112 square units
- D) 192 square units

217. A circle has circumference 31.4 centimeters. Using 3.14 for π , what is its radius?

- A) 2.5 centimeters
- B) 10 centimeters
- C) 5 centimeters
- D) 15.7 centimeters

218. A cyclist travels 18 miles in 1.5 hours. At the same rate, how far will she travel in 50 minutes?

- A) 9 miles
- B) 12 miles
- C) 15 miles
- D) 10 miles

219. A package weighs 6 pounds 10 ounces. After an item weighing 1 pound 14 ounces is removed, what is the new weight?

- A) 4 pounds 12 ounces
- B) 4 pounds 6 ounces
- C) 4 pounds 8 ounces
- D) 5 pounds 12 ounces

220. The numbers 6, 8, 8, 11, and k have a median of 8 and a range of 9. Which could be k ?

- A) -1
- B) 15
- C) 7
- D) 20

221. A spinner has 5 equal sections numbered 1–5. It is spun twice. What is the probability that both results are even?

- A) $\frac{2}{5}$
- B) $\frac{4}{10}$
- C) $\frac{4}{25}$
- D) $\frac{8}{25}$

222. A survey records 12 red, 18 blue, and 10 green choices. If 20 more people respond and half choose green, what fraction of all 60 responses are green?

- A) $\frac{1}{4}$
- B) $\frac{3}{10}$
- C) $\frac{2}{5}$
- D) $\frac{1}{3}$

223. What is the least common multiple of 18 and 30?

- A) 60
- B) 90
- C) 120
- D) 180

224. Which fraction is equivalent to 0.375?

- A) $\frac{3}{8}$
- B) $\frac{1}{4}$
- C) $\frac{5}{12}$
- D) $\frac{7}{20}$

225. What is the value of $2^3 \times 5 - 3^2$?

- A) 17
- B) 37
- C) 49
- D) 31

226. A positive number has a square root between 11 and 12. Which could be the number?

- A) 120
- B) 145
- C) 130
- D) 156

227. Solve $5x + 7 = 3x + 29$.

- A) 9
- B) 18
- C) 11
- D) 22

228. If $y = 2x - 5$ and $x = y + 1$, what is x ?

- A) 2
- B) 4
- C) 3
- D) 6

229. Which expression is equivalent to $3(2a - 5) + 4a$?

- A) $10a - 15$
- B) $6a - 1$
- C) $10a - 5$
- D) $14a - 15$

230. The product of two consecutive positive integers is 72. What is their sum?

- A) 15
- B) 16
- C) 18
- D) 17

231. Two parallel lines are cut by a transversal. One acute angle measures 37° . What is the measure of each obtuse angle?

- A) 143°
- B) 53°
- C) 74°
- D) 127°

232. A right triangle has legs of 9 and 12 units. What is the hypotenuse?

- A) 13 units
- B) 15 units
- C) 18 units
- D) 21 units

233. A cylinder has radius 3 and height 7. What is its volume in terms of π ?

- A) 21π cubic units
- B) 42π cubic units
- C) 63π cubic units
- D) 126π cubic units

234. A map scale is 1 inch to 18 miles. Two towns are 3.5 inches apart. What is their actual distance?

- A) 54 miles
- B) 58.5 miles
- C) 72 miles
- D) 63 miles

235. A tank contains 14.6 liters. After 3,750 milliliters are removed, how many liters remain?

- A) 10.85 liters
- B) 10.25 liters
- C) 11.15 liters
- D) 11.85 liters

236. The mean of 14, 19, 22, and x is 20. What is x ?

- A) 20
- B) 25
- C) 23
- D) 27

237. A bag has 4 yellow, 3 purple, and 5 orange counters. What is the probability of selecting a counter that is not purple?

- A) $\frac{1}{4}$
- B) $\frac{5}{12}$
- C) $\frac{3}{4}$
- D) $\frac{2}{3}$

238. The data are 4, 6, 9, 9, 12, 14. If 20 is added, which measure must increase?

- A) median
- B) mode
- C) minimum
- D) mean

Section 5: Language

60 questions | Recommended time: 30 minutes

239. Choose the best response. Which sentence correctly punctuates a direct quotation?

- A) Mina asked, "Is the telescope ready?"
- B) Mina asked "Is the telescope ready"?
- C) Mina asked, "Is the telescope ready".
- D) Mina asked "Is the telescope ready?"

240. Choose the best response. Which sentence correctly joins two closely related independent clauses?

- A) The rain stopped, the championship game continued.
- B) The rain stopped; the championship game continued.
- C) The rain stopped: and the championship game continued.
- D) The rain stopped the championship game; continued.

241. Choose the best response. Which sentence is punctuated correctly?

- A) My brother who lives in Denver, designs bridges.
- B) My brother who, lives in Denver designs bridges.
- C) My brother, who lives in Denver, designs bridges.
- D) My brother who lives in Denver designs, bridges.

242. Choose the best response. Which sentence is capitalized correctly?

- A) Professor chen teaches Biology at Westfield Academy.
- B) professor Chen teaches biology at westfield academy.
- C) Professor Chen teaches Biology at Westfield academy.
- D) Professor Chen teaches biology at Westfield Academy.

243. Choose the best response. Which sentence is capitalized correctly?

- A) We crossed the Pacific Ocean before arriving in New Zealand.
- B) We crossed the pacific ocean before arriving in New Zealand.
- C) We crossed the Pacific ocean before arriving in new zealand.
- D) We crossed the pacific Ocean before arriving in New zealand.

244. Choose the best response. Choose the correctly spelled word meaning "able to be relied upon."

- A) dependable
- B) dependable
- C) depandable
- D) dependabel

245. Choose the best response. Which sentence contains a misspelled word?

- A) The audience remained attentive.
- B) The schedule changed unexpectedly.
- C) The speaker gave a brief recomendation.
- D) The volunteers organized the supplies.

246. Choose the best response. Which sentence has correct subject-verb agreement?

- A) Neither the drums nor the piano are available.
- B) Neither the drums nor the piano were available now.
- C) Neither the drums nor the piano be available.
- D) Neither the drums nor the piano is available.

247. Choose the best response. Which sentence is complete and correctly formed?

- A) Although the road was steep, the hikers maintained their pace.
- B) Although the road was steep.
- C) The road was steep, the hikers maintained their pace.
- D) Maintaining their pace although the steep road.

248. Choose the best response. Which sentence uses parallel structure correctly?

- A) The internship required filing records, to answer calls, and greeting visitors.
- B) The internship required filing records, answering calls, and greeting visitors.
- C) The internship required records to be filed, answering calls, and visitors were greeted.
- D) The internship required to file records, calls were answered, and greeting visitors.

249. Choose the best response. Which word correctly completes the sentence? The hikers packed _____ water than they needed.

- A) many
- B) muchest
- C) more
- D) most

250. Choose the best response. Which transition best completes the sentence? The first design was inexpensive; _____, it could not support enough weight.

- A) therefore
- B) similarly
- C) for example
- D) however

Editing Passage: A Better Lunch Line

(1) Students at Brookside School often spent most of lunch waiting to buy food. (2) The student council observed the line for one week and found that choosing a meal at the counter caused the longest delays. (3) Council members suggested posting the menu near the entrance so students could decide earlier. (4) The cafeteria serves soup on Thursdays. (5) They also proposed a separate station for students buying only drinks. (6) After the changes, the average wait fell by four minutes. (7) Careful observation and two simple adjustments gave students more time to eat and talk.

251. Choose the best response. Which sentence should be removed because it does not support the passage's main focus?

- A) Sentence 4
- B) Sentence 2
- C) Sentence 5
- D) Sentence 6

252. Choose the best response. Which sentence would best introduce the passage?

- A) School lunches contain many different foods.
- B) Student councils meet throughout the year.
- C) A small investigation helped students solve a frustrating cafeteria problem.
- D) Some students prefer to bring lunch from home.

253. Choose the best response. Which revision best combines sentences 2 and 3?

- A) The council observed the line, and one week was long, with the menu at the entrance.
- B) Counter decisions caused delays, the council suggested a menu and observed them.
- C) Posting the menu, the longest delays were observed by the council near the entrance.
- D) After observing that counter decisions caused the longest delays, council members suggested posting the menu near the entrance.

254. Which sentence correctly uses an apostrophe?

- A) The two fox's tracks crossed near the creek.
- B) The two foxes tracks' crossed near the creek.
- C) The two foxes's tracks crossed near the creek.
- D) The two foxes' tracks crossed near the creek.

255. Which sentence correctly uses a colon?

- A) Bring these items: a flashlight, a map, and water.
- B) Bring: these items a flashlight, a map, and water.
- C) Bring these items, a flashlight: a map, and water.
- D) Bring these: items, a flashlight, a map, and water.

256. Which sentence is capitalized correctly?

- A) In july, uncle Mateo will visit Lake Erie.
- B) In July, Uncle Mateo will visit Lake Erie.
- C) In July, Uncle Mateo will visit lake erie.
- D) In July, uncle Mateo will visit Lake erie.

257. Which sentence is capitalized correctly?

- A) The declaration of independence was signed in Philadelphia.
- B) The Declaration Of Independence was signed in philadelphia.
- C) The Declaration of Independence was signed in Philadelphia.
- D) The Declaration of independence was signed in Philadelphia.

258. Choose the correctly spelled word.

- A) ocassionally
- B) occasionly
- C) occasionaly
- D) occasionally

259. Which sentence contains a misspelled word?

- A) We received a sepearte envelope for each form.
- B) The decision was completely reasonable.
- C) Her explanation was concise.
- D) The equipment remained secure.

260. Choose the correctly spelled word meaning “required.”

- A) neccessary
- B) necessary
- C) nessessary
- D) necessary

261. Which sentence uses pronouns correctly?

- A) The coach gave Mia and I the schedule.
- B) The coach gave she and me the schedule.
- C) The coach gave Mia and me the schedule.
- D) The coach gave Mia and myself the schedule.

262. Which sentence uses the correct verb tense?

- A) By the time the bus arrived, we wait for twenty minutes.
- B) By the time the bus arrived, we will waited for twenty minutes.
- C) By the time the bus arrived, we have wait for twenty minutes.
- D) By the time the bus arrived, we had waited for twenty minutes.

263. Which sentence contains no misplaced or dangling modifier?

- A) Walking through the garden, Priya noticed the first tulips.
- B) Walking through the garden, the first tulips appeared to Priya.
- C) Covered in dew, Priya photographed the grass.
- D) After reading the article, the experiment seemed clearer.

264. Which word correctly completes the sentence? The storm did not ____ our travel plans.

- A) effect
- B) affect
- C) effects
- D) affected

265. Which sentence uses the comparative form correctly?

- A) This route is least crowded than the highway.
- B) This route is more less crowded than the highway.
- C) This route is less crowded than the highway.
- D) This route is lesser crowded than the highway.

Editing Passage: A Garden for Rain

(1) Last spring, our science club transformed a low patch beside the parking lot into a rain garden. (2) During storms, water had collected there and carried soil toward a drain. (3) We loosened the ground, added compost, and planted native grasses whose roots absorb water. (4) Native plants grow in many regions. (5) After several months, the garden held runoff long enough for much of it to soak into the soil. (6) The project reduced erosion, and it also created shelter for insects. (7) By studying one troublesome spot, our club found a solution with several benefits.

266. Which sentence should be removed because it is too general to support the passage's focus?

- A) Sentence 2
- B) Sentence 5
- C) Sentence 6
- D) Sentence 4

267. Which revision of sentence 6 is most concise?

- A) The project reduced erosion and created insect shelter.
- B) The project reduced erosion, and insect shelter was also something it created.
- C) Reducing erosion was done by the project, and it created shelter too.
- D) The project was a project that reduced erosion and also created shelter for insects.

268. Which sentence would best follow sentence 3?

- A) The club meets every other Wednesday after school.
- B) We chose species that could tolerate both temporary flooding and dry summer days.
- C) Parking lots contain painted spaces for cars.
- D) Some students prefer vegetables to grasses.

269. Which sentence is punctuated correctly?

- A) Before sunrise, we packed the tents, folded the blankets, and swept the cabin.
- B) Before sunrise we packed, the tents folded the blankets and swept the cabin.
- C) Before sunrise; we packed the tents, folded the blankets and, swept the cabin.
- D) Before sunrise we packed the tents folded the blankets, and swept the cabin.

270. Which sentence correctly punctuates a quotation?

- A) "Please label every sample", Dr. Chen said.
- B) "Please label every sample," Dr. Chen said.
- C) "Please label every sample" Dr. Chen, said.
- D) "Please label every sample." Dr. Chen said.

271. Which sentence correctly uses a semicolon?

- A) The trail was muddy, nevertheless; the group continued.
- B) The trail; was muddy nevertheless, the group continued.
- C) The trail was muddy; nevertheless, the group continued.
- D) The trail was muddy nevertheless; the group continued.

272. Which option correctly capitalizes the poem title and course name?

- A) Our Class read "The Road Not Taken" in english class.
- B) Our class read "The road not taken" in English Class.
- C) Our class read "the Road Not Taken" in english class.
- D) Our class read "The Road Not Taken" in English class.

273. Which option correctly capitalizes the title, place name, and holiday?

- A) The mayor spoke at Central Library on Memorial Day.
- B) The Mayor spoke at central library on Memorial Day.
- C) The mayor spoke at Central Library on memorial day.
- D) The mayor spoke at central Library on Memorial day.

274. Which option correctly spells the word meaning “a special right”?

- A) privelege
- B) privilege
- C) priviledge
- D) privilage

275. In which sentence is one word spelled incorrectly?

- A) The audience remained attentive.
- B) The boundary was clearly marked.
- C) The committee will recomend a new schedule.
- D) The change occurred gradually.

276. Which sentence has correct subject-verb agreement?

- A) Neither the lamps nor the clock belong in this box.
- B) Neither the lamps nor the clock are belonging in this box.
- C) Neither the lamps nor the clock have belonged in this box.
- D) Neither the lamps nor the clock belongs in this box.

277. Which sentence uses parallel structure?

- A) The internship required filing records, answering calls, and preparing reports.
- B) The internship required filing records, to answer calls, and reports were prepared.
- C) The internship required records to be filed, answering calls, and report preparation.
- D) The internship required to file records, calls were answered, and preparing reports.

278. Which sentence correctly combines the ideas? The rain stopped. The players returned to the field.

- A) The rain stopped, the players returned to the field.
- B) When the rain stopped, the players returned to the field.
- C) Stopping the rain, the field returned the players.
- D) The players returned, and because to the field the rain stopped.

279. Which word correctly completes the sentence? The new evidence may ____ the committee’s decision.

- A) altar
- B) allude
- C) alter
- D) adapt

280. Which sentence uses fewer and less correctly?

- A) Less cars meant fewer noise near the school.
- B) Fewer cars meant fewer noise near the school.
- C) Less cars meant less noise near the school.
- D) Fewer cars meant less noise near the school.

281. Which sentence uses the word between correctly?

- A) The treaty settled the dispute between the two towns.
- B) The treaty settled the dispute among the two towns.
- C) The treaty settled the dispute beside the two towns.
- D) The treaty settled the dispute within of the two towns.

Editing Passage: The Night Survey

(1) Last October, our environmental club joined a nighttime survey of moths. (2) A biologist hung a white sheet between two trees and aimed a safe light toward it. (3) Moths landed on the cloth, where we photographed their wing patterns and recorded their sizes. (4) We worked quietly so that sudden movement would not disturb them. (5) In two hours, the group documented seventeen species. (6) The night was cold. (7) Before leaving, we checked the area for equipment and uploaded our observations to a shared database. (8) The survey showed us how a simple, careful method can reveal animals that usually go unnoticed.

282. Which sentence should be removed because it does not contribute meaningfully to the passage?

- A) Sentence 2
- B) Sentence 6
- C) Sentence 5
- D) Sentence 8

283. Which sentence would best follow sentence 5?

- A) The park also has benches near its entrance.
- B) Seventeen is a number greater than sixteen.
- C) Several had never before been recorded in that section of the park.
- D) Our school cafeteria serves lunch at noon.

284. Which option correctly punctuates the introductory clause and series?

- A) After the rehearsal ended Maya, returned her costume thanked the director, and called her father.
- B) After the rehearsal ended; Maya returned her costume, thanked the director and called, her father.
- C) After the rehearsal ended, Maya returned her costume, thanked the director, and called her father.
- D) After the rehearsal ended Maya returned her costume thanked the director and, called her father.

285. Which sentence correctly shows possession?

- A) The two coaches' offices are beside the gym.
- B) The two coaches office's are beside the gym.
- C) The two coach's offices are beside the gym.
- D) The two coaches offices are beside the gym.

286. Which option correctly spells the word meaning "an event or instance"?

- A) occurence
- B) occurrence
- C) ocurrence
- D) occurrence

287. Which option correctly capitalizes a month, family title, name, and region?

- A) In july, aunt Rosa will visit the Pacific northwest.
- B) In July, Aunt Rosa will visit the Pacific Northwest.
- C) In July, Aunt rosa will visit the pacific Northwest.
- D) In july, Aunt Rosa will visit the Pacific Northwest.

288. Select the sentence with correct capitalization.

- A) professor Malik teaches History at Westlake college.
- B) Professor Malik teaches History at westlake College.
- C) Professor Malik teaches history at Westlake College.
- D) Professor malik teaches history at Westlake College.

289. Which option correctly spells the word meaning “needed or required”?

- A) necessary
- B) neccessary
- C) necessery
- D) nesessary

290. Find the sentence in which the spelling error appears.

- A) Her explanation was reasonable.
- B) The equipment arrived separately.
- C) We admired the unusual architecture.
- D) The sudden noise caused a brief interruption.

291. Which sentence uses the pronoun correctly?

- A) The librarian gave Lena and I the forms.
- B) The librarian gave Lena and me the forms.
- C) The librarian gave she and me the forms.
- D) The librarian gave Lena and myself the forms.

292. Select the sentence whose opening modifier clearly describes its subject.

- A) Walking toward the station, the first train arrived.
- B) Covered in frost, Mia admired the window.
- C) Walking toward the station, I saw the first train arrive.
- D) After reading the book, the movie seemed confusing.

293. Which sentence correctly uses verb tense?

- A) By the time the bus arrived, we had waited for thirty minutes.
- B) By the time the bus arrived, we wait for thirty minutes.
- C) By the time the bus arrives, we had waited for thirty minutes yesterday.
- D) By the time the bus arrived, we have wait for thirty minutes.

294. Which word correctly completes the sentence? The speaker tried to ____ the importance of daily practice.

- A) sympathize
- B) summarize
- C) advertise
- D) emphasize

295. Which sentence uses who or whom correctly?

- A) Who did the committee select as its representative?
- B) Whom did the committee select as its representative?
- C) Whom will present the report tomorrow?
- D) The student whom won will present the report.

296. Which sentence uses imply and infer correctly?

- A) The clues infer a delay, and readers may imply its cause.
- B) The clues imply a delay, and readers may imply its cause.
- C) The clues imply a delay, and readers may infer its cause.
- D) The clues infer a delay, and readers may infer its cause.

Editing Passage: The Courtyard Map

(1) Our class decided to create a map of the school courtyard for new students. (2) First, pairs measured the walkways and recorded the locations of benches, doors, and garden beds. (3) Another group designed simple symbols for each feature. (4) I like maps with bright colors. (5) When we combined our work, however, several measurements did not match. (6) We returned to the courtyard, checked the disputed distances, and corrected two recording errors. (7) Finally, we added a key and tested the map by asking another class to follow it. (8) Their successful trip from the library door to the garden confirmed that our revised map was clear.

297. Which sentence should be removed because it distracts from the passage's focus?

- A) Sentence 4
- B) Sentence 2
- C) Sentence 6
- D) Sentence 8

298. Which transition best replaces however in sentence 5?

- A) Instead
- B) For example
- C) Nevertheless
- D) When

Answer Key

Question numbers continue across all five sections.

Verbal Skills

1. A	2. B	3. C	4. D	5. A	6. B	7. C	8. D
9. A	10. B	11. C	12. D	13. A	14. B	15. C	16. B
17. C	18. D	19. A	20. B	21. C	22. D	23. A	24. B
25. C	26. D	27. A	28. B	29. C	30. D	31. A	32. C
33. D	34. A	35. B	36. C	37. D	38. A	39. C	40. D
41. B	42. A	43. C	44. D	45. B	46. B	47. A	48. C
49. D	50. B	51. A	52. C	53. D	54. B	55. C	56. D
57. A	58. C	59. D	60. B				

Quantitative Skills

61. A	62. B	63. C	64. D	65. A	66. B	67. B	68. C
69. A	70. D	71. C	72. B	73. D	74. A	75. C	76. D
77. A	78. C	79. D	80. A	81. B	82. B	83. C	84. B
85. C	86. D	87. A	88. B	89. C	90. D	91. A	92. D
93. C	94. D	95. A	96. C	97. B	98. D	99. B	100. A
101. C	102. D	103. A	104. C	105. D	106. A	107. B	108. C
109. D	110. A	111. B	112. B				

Reading Comprehension

113. A	114. B	115. C	116. D	117. A	118. B	119. C	120. D
121. A	122. B	123. C	124. D	125. A	126. B	127. C	128. D
129. D	130. A	131. B	132. C	133. A	134. B	135. C	136. D
137. A	138. B	139. C	140. D	141. A	142. B	143. C	144. D
145. A	146. B	147. C	148. D	149. A	150. B	151. C	152. D
153. A	154. B	155. C	156. D	157. A	158. B	159. D	160. B
161. C	162. D	163. A	164. B	165. C	166. D	167. A	168. B
169. C	170. D	171. A	172. B	173. C	174. D		

Mathematics

175. A	176. B	177. C	178. D	179. A	180. B	181. C	182. D
183. A	184. B	185. C	186. D	187. A	188. B	189. C	190. D
191. B	192. C	193. D	194. A	195. B	196. C	197. D	198. A
199. B	200. C	201. D	202. A	203. B	204. C	205. D	206. A

207. A	208. B	209. C	210. D	211. A	212. B	213. C	214. D
215. A	216. B	217. C	218. D	219. A	220. B	221. C	222. D
223. B	224. A	225. D	226. C	227. C	228. B	229. A	230. D
231. A	232. B	233. C	234. D	235. A	236. B	237. C	238. D

Language

239. A	240. B	241. C	242. D	243. A	244. B	245. C	246. D
247. A	248. B	249. C	250. D	251. A	252. C	253. D	254. D
255. A	256. B	257. C	258. D	259. A	260. B	261. C	262. D
263. A	264. B	265. C	266. D	267. A	268. B	269. A	270. B
271. C	272. D	273. A	274. B	275. C	276. D	277. A	278. B
279. C	280. D	281. A	282. B	283. C	284. C	285. A	286. D
287. B	288. C	289. A	290. D	291. B	292. C	293. A	294. D
295. B	296. C	297. A	298. D				

Detailed Explanations

Verbal Skills

1. HSPT-V4-001 - Answer A: unwilling

Reluctant means hesitant or unwilling to act.

2. HSPT-V4-002 - Answer B: protect

To preserve something is to protect it from harm or loss.

3. HSPT-V4-003 - Answer C: obvious

Evident means clear or obvious.

4. HSPT-V4-004 - Answer D: plentiful

Scarce means limited in supply; plentiful means abundant.

5. HSPT-V4-005 - Answer A: flexible

Rigid means stiff or unyielding; flexible means able to bend or adapt.

6. HSPT-V4-006 - Answer B: reveal

To conceal is to hide; to reveal is to make known or visible.

7. HSPT-V4-007 - Answer C: meal

A blueprint provides a plan for a building; a recipe provides a plan for a meal.

8. HSPT-V4-008 - Answer D: silkworm

Wool comes from sheep; silk is produced by silkworms.

9. HSPT-V4-009 - Answer A: solid

Evaporation changes a liquid; melting changes a solid.

10. HSPT-V4-010 - Answer B: harbor

Prairie, tundra, and desert are land biomes; a harbor is a sheltered body of water.

11. HSPT-V4-011 - Answer C: ballad

All four are recognized forms of poetry.

12. HSPT-V4-012 - Answer D: hive and bee

A kennel houses a dog; a hive houses a bee colony.

13. HSPT-V4-013 - Answer A: No object in cabinet P is transparent.

Because every cabinet-P object is metal and no workshop metal object is transparent, none of those objects is transparent.

14. HSPT-V4-014 - Answer B: W

The only order satisfying all constraints is Y, X, W, Z.

15. HSPT-V4-015 - Answer C: Code R is not approved.

Ending with K is necessary for approval, so a code lacking that ending cannot be approved.

16. HSPT-V4-016 - Answer B: wise

Prudent means showing careful, wise judgment.

17. HSPT-V4-017 - Answer C: decrease

To diminish is to become or make smaller.

18. HSPT-V4-018 - Answer D: confirm

To verify something is to confirm that it is true or accurate.

19. HSPT-V4-019 - Answer A: simple

Intricate means highly detailed or complicated; simple is the opposite.

20. HSPT-V4-020 - Answer B: forbid

To permit is to allow; to forbid is to prohibit.

21. HSPT-V4-021 - Answer C: pessimistic

Optimistic expects favorable results; pessimistic expects unfavorable ones.

22. HSPT-V4-022 - Answer D: temperature

A compass measures or indicates direction; a thermometer measures temperature.

23. HSPT-V4-023 - Answer A: cow

A sapling is a young tree; a calf is a young cow.

24. HSPT-V4-024 - Answer B: trial
Applause follows and responds to a performance; a verdict concludes a trial.
25. HSPT-V4-025 - Answer C: copper
Granite, marble, and slate are rocks; copper is a metal.
26. HSPT-V4-026 - Answer D: spin
All four words describe circular motion.
27. HSPT-V4-027 - Answer A: scene and play
A chapter is a division of a book; a scene is a division of a play.
28. HSPT-V4-028 - Answer B: Ceramics is offered Tuesday.
Exactly one day is used, so excluding Thursday forces Tuesday.
29. HSPT-V4-029 - Answer C: measuring
Kira sorts; Luis must paint, leaving measuring for Jules.
30. HSPT-V4-030 - Answer D: The gate is not locked.
The contrapositive of the rule is: if the light is off, the gate is not locked.
31. HSPT-V4-031 - Answer A: unwilling
Reluctant means hesitant or unwilling.
32. HSPT-V4-032 - Answer C: limited
Scarce means in short or limited supply.
33. HSPT-V4-033 - Answer D: detailed
Something elaborate contains many careful details.
34. HSPT-V4-034 - Answer A: flexible
Rigid means stiff; flexible is its opposite.
35. HSPT-V4-035 - Answer B: reveal
To conceal is to hide; to reveal is to make known.
36. HSPT-V4-036 - Answer C: meager
Abundant means plentiful; meager means insufficient or scant.
37. HSPT-V4-037 - Answer D: grain
A reservoir stores water; a granary stores grain.
38. HSPT-V4-038 - Answer A: physician
A sculptor uses a chisel; a physician uses a stethoscope.
39. HSPT-V4-039 - Answer C: walk
A whisper is a gentle way to speak; a stroll is a leisurely way to walk.
40. HSPT-V4-040 - Answer D: harbor
A lagoon, pond, and lake are bodies of mostly enclosed water; a harbor is a sheltered place for vessels.
41. HSPT-V4-041 - Answer B: pine
Cedar, spruce, fir, and pine are evergreen coniferous trees.
42. HSPT-V4-042 - Answer A: wolf and pack
A kitten may belong to a litter; a wolf may belong to a pack.
43. HSPT-V4-043 - Answer C: No manuscript in the cabinet may leave the archive.
The two rules place every cabinet manuscript among items that cannot leave.
44. HSPT-V4-044 - Answer D: M
The block M-J-K must occur in that order; with L not at an end, only M can be first.
45. HSPT-V4-045 - Answer B: Bo and Cy
Bo and Cy satisfies exactly two attendees and does not violate either conditional rule.
46. HSPT-V4-046 - Answer B: watchful
Vigilant means alert and watchful for possible trouble.
47. HSPT-V4-047 - Answer A: fair
Impartial describes someone who is fair and does not favor one side.
48. HSPT-V4-048 - Answer C: possible
Feasible describes something that is possible and practical to accomplish.
49. HSPT-V4-049 - Answer D: permanent
Temporary means lasting a limited time; permanent means lasting indefinitely.

50. HSPT-V4-050 - Answer B: bold
Timid means fearful or lacking confidence; bold expresses its opposite.
51. HSPT-V4-051 - Answer A: assist
To hinder is to obstruct progress; to assist is to help it.
52. HSPT-V4-052 - Answer C: meal
A blueprint guides construction of a building; a recipe guides preparation of a meal.
53. HSPT-V4-053 - Answer D: insect
Fish breathe through gills; many insects breathe through spiracles.
54. HSPT-V4-054 - Answer B: water
Opaque material blocks light; impermeable material blocks water from passing through.
55. HSPT-V4-055 - Answer C: sonnet
Novel, memoir, and biography are prose forms; a sonnet is a poetic form.
56. HSPT-V4-056 - Answer D: tin
Copper, silver, iron, and tin are metals.
57. HSPT-V4-057 - Answer A: apiary and bees
An aviary houses birds; an apiary houses colonies of bees.
58. HSPT-V4-058 - Answer C: No cedar box is stored in Room K.
Because every cedar box is fragrant, the rule excludes every cedar box from Room K.
59. HSPT-V4-059 - Answer D: Oren
The constraints force the unique order Mara, Oren, Pia, Niko.
60. HSPT-V4-060 - Answer B: 426
Only 426 has distinct allowed digits, a first digit twice the second, and a larger final digit.

Quantitative Skills

61. HSPT-Q4-001 - Answer A: 15

The differences are -9 , -12 , -15 , -18 , then -21 ; $36 - 21 = 15$.

62. HSPT-Q4-002 - Answer B: 13.5

The rule alternates $\times 3$ and $\div 2$, so $27 \div 2 = 13.5$.

63. HSPT-Q4-003 - Answer C: 46

Successive increases are 3, 6, 9, and 12; the next increase is 15.

64. HSPT-Q4-004 - Answer D: 27

Add consecutive odd numbers 1, 3, 5, 7, then 9; $18 + 9 = 27$.

65. HSPT-Q4-005 - Answer A: \$85

The sale price is 80% of the original, so $68 \div 0.80 = 85$.

66. HSPT-Q4-006 - Answer B: 61

The number is $7 \times 8 + 5 = 61$.

67. HSPT-Q4-007 - Answer B: 12

There are eight equal parts; $32 \div 8 = 4$, and $3 \times 4 = 12$.

68. HSPT-Q4-008 - Answer C: The two quantities are equal.

$5/6 - 1/4 = 10/12 - 3/12 = 7/12$, so the quantities are equal.

69. HSPT-Q4-009 - Answer A: Quantity A is greater.

For every integer n greater than 4, $3n$ exceeds $n + 8$.

70. HSPT-Q4-010 - Answer D: The relationship cannot be determined from the information given.

Different positive factor pairs produce sums below, equal to, or above 10, so the relationship is undetermined.

71. HSPT-Q4-011 - Answer C: The two quantities are equal.

The radius is half the diameter, so both quantities equal 7.

72. HSPT-Q4-012 - Answer B: Quantity B is greater.

The hexagon's perimeter is $6 \times 4 = 24$, so Quantity B is greater.

73. HSPT-Q4-013 - Answer D: The relationship cannot be determined from the information given.

Without the value of x , the width may be less than, equal to, or greater than 6.

74. HSPT-Q4-014 - Answer A: 10

Each term is half the preceding term.

75. HSPT-Q4-015 - Answer C: 127

Each term is twice the previous term plus 1.

76. HSPT-Q4-016 - Answer D: 16

Two interwoven sequences double: 2, 4, 8, 16 and 5, 10, 20.

77. HSPT-Q4-017 - Answer A: 9

Divide by 5 to get $k - 2 = 7$, so $k = 9$.

78. HSPT-Q4-018 - Answer C: 120

The least common multiple uses $2^3 \times 3 \times 5 = 120$.

79. HSPT-Q4-019 - Answer D: 1 cup

The scale factor is $1/2$ cup per ratio part, so two oat parts equal 1 cup.

80. HSPT-Q4-020 - Answer A: 15

Six percent of 250 is $0.06 \times 250 = 15$.

81. HSPT-Q4-021 - Answer B: Quantity B is greater.

Quantity A is $8 + 9 = 17$, so Quantity B is greater.

82. HSPT-Q4-022 - Answer B: Quantity B is greater.

For every x , $x + 4$ is eight greater than $x - 4$.

83. HSPT-Q4-023 - Answer C: The two quantities are equal.

For every nonzero a , $a/a = 1$.

84. HSPT-Q4-024 - Answer B: Quantity B is greater.

The perimeter is 16, so Quantity B is greater.

85. HSPT-Q4-025 - Answer C: The two quantities are equal.
The width is $48 \div 8 = 6$ units, so the quantities are equal.
86. HSPT-Q4-026 - Answer D: The relationship cannot be determined from the information given.
Comparing $2r$ and $r + 5$ depends on whether r is less than, equal to, or greater than 5.
87. HSPT-Q4-027 - Answer A: 59
The differences are 5, 8, 11, 14; the next difference is 17.
88. HSPT-Q4-028 - Answer B: $13/3$
The operations alternate $\div 3$ and $+ 3$.
89. HSPT-Q4-029 - Answer C: 28
Odd-position terms are 1, 3, 5, 7; each following term is four times it.
90. HSPT-Q4-030 - Answer D: 720
The terms are multiplied successively by 3, 4, 5, and then 6.
91. HSPT-Q4-031 - Answer A: 70
Multiply 42 by $5/3$ to get 70.
92. HSPT-Q4-032 - Answer D: 72
72 is $70 + 2$, so it leaves remainder 2 for both divisors.
93. HSPT-Q4-033 - Answer C: 25%
The increase is 12, and $12/48 = 25\%$.
94. HSPT-Q4-034 - Answer D: The relationship cannot be determined from the information given.
A prime n has two factors, while a composite n may have more; the relationship varies.
95. HSPT-Q4-035 - Answer A: Quantity A is greater.
Quantity A is 11 and Quantity B is 9.
96. HSPT-Q4-036 - Answer C: The two quantities are equal.
Substitution gives Quantity A = $20 - 6 = 14$.
97. HSPT-Q4-037 - Answer B: Quantity B is greater.
The perimeter is $6 \times 4 = 24$, so Quantity B is greater.
98. HSPT-Q4-038 - Answer D: The relationship cannot be determined from the information given.
The other angle is $180 - x$, which may be below, equal to, or above 90.
99. HSPT-Q4-039 - Answer B: Quantity B is greater.
A square side is $10/\sqrt{2}$, about 7.07, so Quantity B is greater.
100. HSPT-Q4-040 - Answer A: 158
Each term is twice the previous term plus 2, so $78 \times 2 + 2 = 158$.
101. HSPT-Q4-041 - Answer C: 15
The operations alternate dividing by 2 and adding 4.
102. HSPT-Q4-042 - Answer D: 40
Two interwoven sequences double: 2, 4, 8, 16 and 5, 10, 20, 40.
103. HSPT-Q4-043 - Answer A: 52
Successive differences are 3, 6, 9, 12, then 15; $37 + 15 = 52$.
104. HSPT-Q4-044 - Answer C: 72
If $1.25n = 90$, then $n = 90 \div 1.25 = 72$.
105. HSPT-Q4-045 - Answer D: 53
The integer equals $8 \times 6 + 5 = 53$.
106. HSPT-Q4-046 - Answer A: 24
There are 8 ratio parts; each is 8 tiles, so white tiles total $3 \times 8 = 24$.
107. HSPT-Q4-047 - Answer B: Quantity B is greater.
A negative number cubed is negative, while its square is positive, so Quantity B is greater.
108. HSPT-Q4-048 - Answer C: The two quantities are equal.
Since $a = b + 9$, Quantity A equals $b + 13$, the same as Quantity B.
109. HSPT-Q4-049 - Answer D: The relationship cannot be determined from the information given.
Possible factor pairs give sums 25, 14, 11, or 10, so the relationship varies.
110. HSPT-Q4-050 - Answer A: Quantity A is greater.
The area is 45, while the perimeter is $2(9 + 5) = 28$.

111. HSPT-Q4-051 - Answer B: Quantity B is greater.

The third angle is $180 - 136 = 44^\circ$, so Quantity B is greater.

112. HSPT-Q4-052 - Answer B: Quantity B is greater.

The volume is 27 and the surface area is $6 \times 9 = 54$, so Quantity B is greater.

Reading Comprehension

113. HSPT-R4-001 - Answer A: Her moving boxes are nearly full, and the new studio has less storage.
The passage directly gives limited box space and a smaller studio as the reasons.
114. HSPT-R4-002 - Answer B: He thinks of them as completed memories rather than reusable material.
Nico values each form as a keepsake, while his aunt sees clay that can change.
115. HSPT-R4-003 - Answer C: unglazed
The plain whistles are contrasted with glazed pieces from which glaze must be removed.
116. HSPT-R4-004 - Answer D: the possibility of creating something new from the past
By the ending, Nico sees the old clay as material for future creations.
117. HSPT-R4-005 - Answer A: Letting an object change can preserve its value in a new form.
The whistles are not simply discarded; their clay gains future possibilities through reuse.
118. HSPT-R4-006 - Answer B: Moisture, oxygen, and suitable temperature
The first paragraph names moisture, oxygen, and suitable temperature.
119. HSPT-R4-007 - Answer C: It prevents sprouting during a brief autumn warm spell.
The cold requirement delays sprouting until after winter.
120. HSPT-R4-008 - Answer D: They face more sunlight and nutrients with less competition.
The passage says fire opens sunny space and returns nutrients to soil.
121. HSPT-R4-009 - Answer A: signals
Environmental cues are conditions that signal when a seed should begin growing.
122. HSPT-R4-010 - Answer B: It explains dormancy, gives examples of triggers, and states its survival advantage.
The passage moves from the concept to examples and then to its overall function.
123. HSPT-R4-011 - Answer C: Weaknesses before costly materials are used
The third paragraph explicitly identifies this benefit of an early model.
124. HSPT-R4-012 - Answer D: Unfinished work is valuable when it makes problems visible for thoughtful revision.
All examples support using an imperfect version to guide purposeful improvement.
125. HSPT-R4-013 - Answer A: It may show that a writer evaluated and improved an idea.
The author calls crossed-out sentences records of decisions, not evidence of failure.
126. HSPT-R4-014 - Answer B: analogies extending the idea beyond writing
Both examples show how preliminary versions permit correction in other fields.
127. HSPT-R4-015 - Answer C: reasoned
The author carefully qualifies why drafts help and distinguishes revision from repetition.
128. HSPT-R4-016 - Answer D: insufficient
Thin evidence is evidence that is too limited to support the writing well.
129. HSPT-R4-017 - Answer D: She rushes the opening and misses a shift.
The opening paragraph directly identifies her rushed entrance and missed shift.
130. HSPT-R4-018 - Answer A: He wants her to learn how sound behaves in the auditorium.
The exercise helps Mara listen to how the hall carries and reflects her sound.
131. HSPT-R4-019 - Answer B: metaphor
The hall is figuratively presented as something Mara can learn to play.
132. HSPT-R4-020 - Answer C: Careful attention can turn an unfamiliar setting and a mistake into useful guidance.
Mara uses her error and the hall's acoustics to adjust her performance.
133. HSPT-R4-021 - Answer A: She first sees it as unsettling but later understands how to work with it.
The hall changes from an unfamiliar threat into a useful part of performance.
134. HSPT-R4-022 - Answer B: the distance to an object
The first paragraph states that echo delay indicates distance.
135. HSPT-R4-023 - Answer C: distinguish
The bat must distinguish useful echoes from competing sounds.
136. HSPT-R4-024 - Answer D: They need frequent updates about a nearby moving target.
Shorter intervals provide updated location information during pursuit.

137. HSPT-R4-025 - Answer A: How bats actively adjust echolocation to gather information in different conditions
The passage explains the signals, processing, and adjustments involved in echolocation.
138. HSPT-R4-026 - Answer B: To introduce a familiar comparison and then clarify its limitation
The comparison helps readers, but the author emphasizes echolocation's active feedback process.
139. HSPT-R4-027 - Answer C: Old maps, drainage patterns, soil, and nearby structures
The second paragraph explicitly lists these sources of information.
140. HSPT-R4-028 - Answer D: Modern buildings and utilities may occupy the original path.
Existing urban structures can make the former route impractical.
141. HSPT-R4-029 - Answer A: They can slow runoff instead of sending it away immediately.
The passage says these features slow runoff and temporarily hold water.
142. HSPT-R4-030 - Answer B: land along the sides of a stream
The banks are planted areas beside the restored channel that may erode.
143. HSPT-R4-031 - Answer C: Daylighting adapts an old waterway to modern conditions and requires careful design.
The passage balances potential benefits with modern constraints and risks.
144. HSPT-R4-032 - Answer D: It describes a past practice, explains a redesign process, and weighs benefits and challenges.
The paragraphs move from burial to restoration methods, effects, and a qualified conclusion.
145. HSPT-R4-033 - Answer A: The display case has a broken latch.
The second paragraph states that the broken latch makes the display unsafe.
146. HSPT-R4-034 - Answer B: She wants Nico to use the bell to gather arriving visitors.
The electric chime has failed and the visitors need a signal to gather.
147. HSPT-R4-035 - Answer C: It connects the building's history with its reopening.
The historic object becomes useful in the present at the restored meetinghouse.
148. HSPT-R4-036 - Answer D: became quieter
The visitors' conversations become quieter when the bell rings.
149. HSPT-R4-037 - Answer A: An old object can gain fresh meaning when it serves a present need.
The bell moves from imagined history to renewed use during the reopening.
150. HSPT-R4-038 - Answer B: The left wheel is slightly larger than the right.
Dev measures the wheels and finds stretched tread on the left one.
151. HSPT-R4-039 - Answer C: She believes her code should compensate for imperfect equipment.
The passage directly contrasts her confidence in the program with the hardware issue.
152. HSPT-R4-040 - Answer D: A repeated failure leads to one discovery, then another, and finally a broader lesson.
The team diagnoses wheel size, encounters loose tape, and expands its understanding of the system.
153. HSPT-R4-041 - Answer A: Tessa's embarrassment about the failures
The figurative description presents each visible failure as criticism Tessa feels everyone can see.
154. HSPT-R4-042 - Answer B: Successful design requires considering software, hardware, and surroundings together.
Their final note states that a system includes everything on which it depends.
155. HSPT-R4-043 - Answer C: It can make the wood expand and contract.
The first paragraph identifies expansion and contraction as effects of humidity changes.
156. HSPT-R4-044 - Answer D: easily damaged
The objects require limited exposure and controlled storage because they are vulnerable to damage.
157. HSPT-R4-045 - Answer A: Its meaning can change with the question and surrounding objects.
The passage explains that context allows an object to help tell several stories.
158. HSPT-R4-046 - Answer B: Why keeping some museum objects off display can preserve them and support future use
The passage presents preservation, research, access, and future-exhibit reasons for rotation.
159. HSPT-R4-047 - Answer D: The author corrects a possible misconception by explaining several purposes of storage.
The passage counters the idea that stored objects are forgotten and explains their continuing roles.
160. HSPT-R4-048 - Answer B: She thinks visitors created an improper shortcut.
The first paragraph says she assumes impatient visitors created the shortcut.
161. HSPT-R4-049 - Answer C: To help people notice the route without obstructing animals
The stones make the crossing visible so people can avoid stepping on it.

162. HSPT-R4-050 - Answer D: The needs and activity that humans may overlook
Discovering who uses the trail broadens Amaya's view beyond human convenience.
163. HSPT-R4-051 - Answer A: ease or usefulness
Convenience refers to what makes movement easier or more useful for people.
164. HSPT-R4-052 - Answer B: Careful observation can challenge a quick assumption.
Amaya observes wildlife and revises her original conclusion about the strip.
165. HSPT-R4-053 - Answer C: He has one cup less honey than he needs.
The opening paragraph identifies a one-cup shortage of honey.
166. HSPT-R4-054 - Answer D: He wants the name to honestly reflect his altered drink.
Because the result differs greatly from the original, he gives it an accurate new name.
167. HSPT-R4-055 - Answer A: It creates a problem that leads Eli to invent a new recipe.
The missing honey drives Eli to use peaches and create a successful variation.
168. HSPT-R4-056 - Answer B: The shortage unexpectedly inspired a creative solution.
Her wordplay reframes an obstacle as the source of Eli's invention.
169. HSPT-R4-057 - Answer C: Adapting honestly can turn a limitation into an opportunity.
Eli acknowledges his change and successfully transforms a shortage into something new.
170. HSPT-R4-058 - Answer D: Reducing moisture slows the processes that age them.
The second paragraph directly connects lower moisture with slower chemical activity and aging.
171. HSPT-R4-059 - Answer A: begin to grow
The passage tests whether stored seeds still sprout, or begin growing.
172. HSPT-R4-060 - Answer B: The origin may help researchers understand its traits and conditions.
Collection data gives useful context for studying and using a sample.
173. HSPT-R4-061 - Answer C: How seed banks store, maintain, and make plant diversity available
The passage explains seed-bank procedures, limits, and value as a backup resource.
174. HSPT-R4-062 - Answer D: To clarify both the value and the limits of stored collections
The distinction explains that storage provides backup while living populations continue adapting.

Mathematics

175. HSPT-M4-001 - Answer A: 9

$|-17| = 17$ and $|-8| = 8$, so $17 - 8 = 9$.

176. HSPT-M4-002 - Answer B: $11/15$

$11/15$ is about 0.733, greater than 0.72, 0.71, and 0.70.

177. HSPT-M4-003 - Answer C: 42

The common prime factors produce $2 \times 3 \times 7 = 42$.

178. HSPT-M4-004 - Answer D: 60 ounces

The added 18 ounces equals $3/10$ of capacity, so capacity is $18 \div 0.3 = 60$.

179. HSPT-M4-005 - Answer A: 5

Simplify to $13 - 6x = -17$; then $-6x = -30$ and $x = 5$.

180. HSPT-M4-006 - Answer B: 10

Substitute -2 : $(-2)^2 - 3(-2) = 4 + 6 = 10$.

181. HSPT-M4-007 - Answer C: 6

Set $-5 = -2x + 7$; subtract 7 and divide by -2 to get $x = 6$.

182. HSPT-M4-008 - Answer D: 6

With length $2w + 3$, $2[(2w + 3) + w] = 42$ gives $6w + 6 = 42$ and $w = 6$.

183. HSPT-M4-009 - Answer A: 8

Exterior angles total 360° , and $360 \div 45 = 8$.

184. HSPT-M4-010 - Answer B: 10 units

Since $C = \pi d$, $d = 31.4 \div 3.14 = 10$.

185. HSPT-M4-011 - Answer C: 136 square units

Surface area is $2(3 \times 4 + 3 \times 8 + 4 \times 8) = 2(68) = 136$.

186. HSPT-M4-012 - Answer D: 15 miles per hour

Seventy-two minutes is 1.2 hours, and $18 \div 1.2 = 15$ miles per hour.

187. HSPT-M4-013 - Answer A: 2,350 grams

One kilogram equals 1,000 grams, so $2.35 \times 1,000 = 2,350$.

188. HSPT-M4-014 - Answer B: 3

The range is $14 - 4 = 10$ and the mode is 7, so $10 - 7 = 3$.

189. HSPT-M4-015 - Answer C: $1/2$

The prime outcomes are 2, 3, 5, and 7: four of eight, or $1/2$.

190. HSPT-M4-016 - Answer D: $1/2$

There are 15 bus-or-bicycle responses among 30 total, so the probability is $15/30 = 1/2$.

191. HSPT-M4-017 - Answer B: $2 \frac{1}{2}$

Convert to $11/4 \div 11/10 = 11/4 \times 10/11 = 5/2$.

192. HSPT-M4-018 - Answer C: $3(18 - 6)$

Only $3(12)$ equals 36; the others equal 30, 31, and 30.

193. HSPT-M4-019 - Answer D: 21

First find $0.35 \times 240 = 84$; then $0.25 \times 84 = 21$.

194. HSPT-M4-020 - Answer A: $3y - 12$

Distribute and combine like terms: $8y - 12 - 5y = 3y - 12$.

195. HSPT-M4-021 - Answer B: 19

Multiply by 3 to get $x + 5 = 24$, then subtract 5.

196. HSPT-M4-022 - Answer C: $2n - 7 \leq 15$

Twice n is $2n$, seven less is $2n - 7$, and at most means $\leq$.

197. HSPT-M4-023 - Answer D: $4 + 2.5m$

The fixed fee is 4 and the mileage charge is 2.5 times m .

198. HSPT-M4-024 - Answer A: 75°

Triangle angles total 180° , so $180 - 38 - 67 = 75$.

199. HSPT-M4-025 - Answer B: 15
By the Pythagorean theorem, $\sqrt{9^2 + 12^2} = \sqrt{225} = 15$.
200. HSPT-M4-026 - Answer C: 77 square units
A parallelogram's area is base $\times$ height: $11 \times 7 = 77$.
201. HSPT-M4-027 - Answer D: 63 miles
Multiply 3.5 inches by 18 miles per inch to get 63 miles.
202. HSPT-M4-028 - Answer A: 22 quarts
Five gallons equal 20 quarts; adding 2 gives 22.
203. HSPT-M4-029 - Answer B: 9:13 p.m.
Adding 2 hours gives 8:47; adding 26 minutes gives 9:13.
204. HSPT-M4-030 - Answer C: 85
The total is 410; the known scores total 325, so the missing score is 85.
205. HSPT-M4-031 - Answer D: $7/12$
Seven of the twelve tiles are green or yellow, so the probability is $7/12$.
206. HSPT-M4-032 - Answer A: 3
Data set B has range $24 - 9 = 15$, which is 3 greater than 12.
207. HSPT-M4-033 - Answer A: $3/4$
The average is $(2/3 + 5/6) \div 2 = (4/6 + 5/6) \div 2 = 3/4$.
208. HSPT-M4-034 - Answer B: 42
The shared prime factors produce $2 \times 3 \times 7 = 42$.
209. HSPT-M4-035 - Answer C: $7/12$
The values are 0.57, about 0.5833, 0.58, and 0.579; $7/12$ is greatest.
210. HSPT-M4-036 - Answer D: 9
Compute parentheses, then division: $6 + 24 \div 6 - 1 = 9$.
211. HSPT-M4-037 - Answer A: 10
Substitute -2: $3(4) - 2 = 10$.
212. HSPT-M4-038 - Answer B: 11
Expand to $4x - 12 = 2x + 10$, so $2x = 22$ and $x = 11$.
213. HSPT-M4-039 - Answer C: $4n + 3$
For $n = 1$, $4n + 3$ gives 7 and increases by 4 per term.
214. HSPT-M4-040 - Answer D: 15
Solve $90 + 18p \leq 360$: $18p \leq 270$, so $p \leq 15$.
215. HSPT-M4-041 - Answer A: 8
Exterior angles total 360° , and $360 \div 45 = 8$.
216. HSPT-M4-042 - Answer B: 136 square units
Surface area is $2(3 \times 4 + 3 \times 8 + 4 \times 8) = 2(68) = 136$.
217. HSPT-M4-043 - Answer C: 5 centimeters
Since $C = 2\pi r$, $r = 31.4 \div 6.28 = 5$.
218. HSPT-M4-044 - Answer D: 10 miles
The rate is 12 miles per hour; 50 minutes is $5/6$ hour, giving 10 miles.
219. HSPT-M4-045 - Answer A: 4 pounds 12 ounces
Borrow 16 ounces: $5 \text{ lb } 26 \text{ oz} - 1 \text{ lb } 14 \text{ oz} = 4 \text{ lb } 12 \text{ oz}$.
220. HSPT-M4-046 - Answer B: 15
With $k = 15$, the median remains 8 and the range is $15 - 6 = 9$.
221. HSPT-M4-047 - Answer C: $4/25$
Each spin has probability $2/5$ of even, so $(2/5)(2/5) = 4/25$.
222. HSPT-M4-048 - Answer D: $1/3$
Green responses become 20 out of 60, which simplifies to $1/3$.
223. HSPT-M4-049 - Answer B: 90
The prime factors needed are $2 \times 3^2 \times 5 = 90$.
224. HSPT-M4-050 - Answer A: $3/8$
 $0.375 = 375/1000$, which simplifies to $3/8$.

225. HSPT-M4-051 - Answer D: 31
Evaluate exponents first: $8 \times 5 - 9 = 31$.
226. HSPT-M4-052 - Answer C: 130
Because $121 < 130 < 144$, its square root lies between 11 and 12.
227. HSPT-M4-053 - Answer C: 11
Subtract $3x$ and 7 to get $2x = 22$, so $x = 11$.
228. HSPT-M4-054 - Answer B: 4
Substitute $y = x - 1$ into $y = 2x - 5$: $x - 1 = 2x - 5$, so $x = 4$.
229. HSPT-M4-055 - Answer A: $10a - 15$
Distribute and combine like terms: $6a - 15 + 4a = 10a - 15$.
230. HSPT-M4-056 - Answer D: 17
The consecutive factors are 8 and 9, whose sum is 17.
231. HSPT-M4-057 - Answer A: 143°
Each obtuse angle is supplementary to 37° , so it measures $180^\circ - 37^\circ = 143^\circ$.
232. HSPT-M4-058 - Answer B: 15 units
The Pythagorean theorem gives $\sqrt{9^2 + 12^2} = \sqrt{225} = 15$.
233. HSPT-M4-059 - Answer C: 63π cubic units
Volume is $\pi r^2 h = \pi(3^2)(7) = 63\pi$.
234. HSPT-M4-060 - Answer D: 63 miles
Multiply 3.5 inches by 18 miles per inch to get 63 miles.
235. HSPT-M4-061 - Answer A: 10.85 liters
3,750 milliliters equals 3.75 liters, and $14.6 - 3.75 = 10.85$.
236. HSPT-M4-062 - Answer B: 25
A mean of 20 requires a total of 80; the known numbers total 55, so $x = 25$.
237. HSPT-M4-063 - Answer C: $3/4$
Nine of 12 counters are not purple, so the probability is $9/12 = 3/4$.
238. HSPT-M4-064 - Answer D: mean
Adding a value above the old mean raises the mean; the median, mode, and minimum remain unchanged.

Language

239. HSPT-L4-001 - Answer A: Mina asked, "Is the telescope ready?"
A comma introduces the quotation, and the question mark belongs inside the closing quotation mark.
240. HSPT-L4-002 - Answer B: The rain stopped; the championship game continued.
A semicolon correctly joins two related independent clauses without a conjunction.
241. HSPT-L4-003 - Answer C: My brother, who lives in Denver, designs bridges.
Commas correctly set off the nonessential clause who lives in Denver.
242. HSPT-L4-004 - Answer D: Professor Chen teaches biology at Westfield Academy.
The title before a name and the school name are capitalized; a general subject is not.
243. HSPT-L4-005 - Answer A: We crossed the Pacific Ocean before arriving in New Zealand.
All significant words in the proper geographic names are capitalized.
244. HSPT-L4-006 - Answer B: dependable
Dependable is the standard spelling.
245. HSPT-L4-007 - Answer C: The speaker gave a brief recommendation.
Recommendation requires two c's and two m's.
246. HSPT-L4-008 - Answer D: Neither the drums nor the piano is available.
With neither...nor, the verb agrees with the nearer singular subject piano.
247. HSPT-L4-009 - Answer A: Although the road was steep, the hikers maintained their pace.
The dependent clause is joined to a complete independent clause.
248. HSPT-L4-010 - Answer B: The internship required filing records, answering calls, and greeting visitors.
All three items use matching gerund forms.
249. HSPT-L4-011 - Answer C: more
More correctly compares the amount packed with the amount needed.
250. HSPT-L4-012 - Answer D: however
However signals the contrast between low cost and inadequate strength.
251. HSPT-L4-013 - Answer A: Sentence 4
The weekly soup detail does not explain the waiting problem, proposed changes, or results.
252. HSPT-L4-014 - Answer C: A small investigation helped students solve a frustrating cafeteria problem.
This topic sentence previews both the problem-solving process and cafeteria setting.
253. HSPT-L4-015 - Answer D: After observing that counter decisions caused the longest delays, council members suggested posting the menu near the entrance.
The revision clearly links the observed cause to the proposed solution without unnecessary repetition.
254. HSPT-L4-016 - Answer D: The two foxes' tracks crossed near the creek.
The plural possessive foxes takes an apostrophe after the s.
255. HSPT-L4-017 - Answer A: Bring these items: a flashlight, a map, and water.
A colon follows a complete clause and introduces the list.
256. HSPT-L4-018 - Answer B: In July, Uncle Mateo will visit Lake Erie.
The month, family title used as a name, and proper lake name are capitalized.
257. HSPT-L4-019 - Answer C: The Declaration of Independence was signed in Philadelphia.
The historical document title and city name use standard capitalization.
258. HSPT-L4-020 - Answer D: occasionally
Occasionally contains two c's, one s, and the ending -ally.
259. HSPT-L4-021 - Answer A: We received a separate envelope for each form.
Separate is spelled with ar in the middle, not er.
260. HSPT-L4-022 - Answer B: necessary
Necessary is spelled with one c and two s's.
261. HSPT-L4-023 - Answer C: The coach gave Mia and me the schedule.
Me is the correct object pronoun after gave.
262. HSPT-L4-024 - Answer D: By the time the bus arrived, we had waited for twenty minutes.
Past perfect correctly shows that the waiting began before the bus arrived.

263. HSPT-L4-025 - Answer A: Walking through the garden, Priya noticed the first tulips.
Priya logically performs the action described by the opening modifier.
264. HSPT-L4-026 - Answer B: affect
Affect is the verb meaning influence.
265. HSPT-L4-027 - Answer C: This route is less crowded than the highway.
Less crowded correctly compares two routes.
266. HSPT-L4-028 - Answer D: Sentence 4
Sentence 4 does not explain this garden's problem, construction, or results.
267. HSPT-L4-029 - Answer A: The project reduced erosion and created insect shelter.
The revision preserves both results without repetition or unnecessary wording.
268. HSPT-L4-030 - Answer B: We chose species that could tolerate both temporary flooding and dry summer days.
This sentence adds a relevant detail about selecting plants for rain-garden conditions.
269. HSPT-L4-031 - Answer A: Before sunrise, we packed the tents, folded the blankets, and swept the cabin.
The introductory phrase and the three items in the series are correctly separated by commas.
270. HSPT-L4-032 - Answer B: "Please label every sample," Dr. Chen said.
A comma belongs inside the closing quotation mark before the dialogue tag.
271. HSPT-L4-033 - Answer C: The trail was muddy; nevertheless, the group continued.
A semicolon joins the independent clauses, and commas set off nevertheless.
272. HSPT-L4-034 - Answer D: Our class read "The Road Not Taken" in English class.
Major words in the poem title and the language-course name English are capitalized.
273. HSPT-L4-035 - Answer A: The mayor spoke at Central Library on Memorial Day.
The official place name and holiday are capitalized; the common title mayor is not.
274. HSPT-L4-036 - Answer B: privilege
Privilege is spelled p-r-i-v-i-l-e-g-e.
275. HSPT-L4-037 - Answer C: The committee will recomend a new schedule.
Recommend requires two c's and two m's.
276. HSPT-L4-038 - Answer D: Neither the lamps nor the clock belongs in this box.
With neither...nor, the verb agrees with the nearer subject clock: belongs.
277. HSPT-L4-039 - Answer A: The internship required filing records, answering calls, and preparing reports.
All three items use matching gerund phrases.
278. HSPT-L4-040 - Answer B: When the rain stopped, the players returned to the field.
The dependent clause clearly shows time and is correctly joined to the main clause.
279. HSPT-L4-041 - Answer C: alter
Alter is the verb meaning change.
280. HSPT-L4-042 - Answer D: Fewer cars meant less noise near the school.
Fewer modifies countable cars; less modifies uncountable noise.
281. HSPT-L4-043 - Answer A: The treaty settled the dispute between the two towns.
Between is standard when referring to two distinct parties.
282. HSPT-L4-044 - Answer B: Sentence 6
Sentence 6 gives an isolated weather detail that is not developed or connected to the survey.
283. HSPT-L4-045 - Answer C: Several had never before been recorded in that section of the park.
The sentence develops the significance of documenting seventeen species.
284. HSPT-L4-046 - Answer C: After the rehearsal ended, Maya returned her costume, thanked the director, and called her father.
A comma follows the introductory clause, and commas correctly separate the series.
285. HSPT-L4-047 - Answer A: The two coaches' offices are beside the gym.
The plural possessive coaches' shows that the offices belong to two coaches.
286. HSPT-L4-048 - Answer D: occurrence
Occurrence is spelled with double c and double r.
287. HSPT-L4-049 - Answer B: In July, Aunt Rosa will visit the Pacific Northwest.
The month, family title used as a name, personal name, and region are capitalized.

288. HSPT-L4-050 - Answer C: Professor Malik teaches history at Westlake College.
The title before a name and the official college name are capitalized; history is not.
289. HSPT-L4-051 - Answer A: necessary
Necessary is spelled n-e-c-e-s-s-a-r-y.
290. HSPT-L4-052 - Answer D: The sudden noise caused a brief interruption.
Interruption requires a double r: interruption.
291. HSPT-L4-053 - Answer B: The librarian gave Lena and me the forms.
Me is the correct object pronoun after the verb gave.
292. HSPT-L4-054 - Answer C: Walking toward the station, I saw the first train arrive.
The introductory modifier logically describes the subject I.
293. HSPT-L4-055 - Answer A: By the time the bus arrived, we had waited for thirty minutes.
Past perfect had waited correctly identifies the action preceding the bus's arrival.
294. HSPT-L4-056 - Answer D: emphasize
Emphasize means to give special importance or attention to something.
295. HSPT-L4-057 - Answer B: Whom did the committee select as its representative?
Whom is correct because it serves as the object of select.
296. HSPT-L4-058 - Answer C: The clues imply a delay, and readers may infer its cause.
A source implies, while a listener or reader infers from evidence.
297. HSPT-L4-059 - Answer A: Sentence 4
Sentence 4 gives an undeveloped personal preference unrelated to the mapping process.
298. HSPT-L4-060 - Answer D: When
When correctly introduces the time at which the mismatched measurements became apparent.